

A General Theory of Real-Time Pushdown Normalization

Semantic Quotients, Locality, Representation Debt, and Causal Scheduling

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Abstract

A deterministic pushdown implementation can preserve a simple semantic state while using a physically complicated stack representation. This paper develops a general theory for separating those two levels in real-time pushdown normalization. The basic object is an exact physical lift $\Theta : C \rightarrow M$ of a deterministic semantic system, together with a chosen canonical section $\sigma : M \rightarrow C$. The induced retraction $N = \sigma\Theta$ identifies semantic fibers and makes representation debt a coordinate-free notion relative to the chosen section: a configuration is debt-free exactly when it lies on that section.

The theory then asks when a semantic transition can be realized directly on the chosen section by a bounded-local LIFO update. For finite-observation tail-injective policies we prove a collapse-capacity bound and an exact three-stage obstruction decomposition: semantic predecessor capacity may already exceed local observation capacity; a chosen section may allocate two same-target semantic sources to the same observation; or, after those tests pass, the local action may still be incompatible with the hidden tail. For ordinary one-top pushdown rewriting the last condition reduces exactly to a suffix/tail criterion.

The same framework yields behavioral safe hulls, semantic containment cores, safe-reset causality, schedule conflict structure, section synthesis, forced but harmless debt, and effective verification. It is instantiated on qualitatively different semantic systems, including weighted counting, a finite regular system, one- and multi-type Dyck semantics, and centered weighted-balance normalization. The examples realize distinct regimes—zero debt, representation-relative obstruction, category-level locality obstruction, forced harmless debt, and latent-fatal debt—showing that semantic uniqueness does not determine a unique physical normalization structure. The weighted-balance and centered $p:q$ families are stress tests rather than the semantic scope of the abstract theory: none of the general theorems assumes an integer residual line or a weighted-balance language. The scope is deliberately narrower than a general theory of all pushdown automata: the results concern exact semantic lifts and bounded-local real-time pushdown normalization policies.

Keywords: deterministic pushdown automata; real-time pushdown normalization; semantic quotient; canonical section; representation debt; LIFO locality; pushdown verification.

1 Introduction

Many deterministic pushdown constructions are easiest to describe operationally. A transition pushes a block, searches through a stack representation, cancels a visible symbol, changes control state, or performs some other local repair. Such descriptions are useful for building a machine, but they obscure a more basic distinction. The information relevant to future language behavior may evolve according to a small deterministic semantic system, while the stack and finite control merely provide

one physical realization of that semantic evolution. Two correct machines may therefore implement exactly the same semantic dynamics while maintaining different stacks, incurring different temporary noncanonical states, and admitting different local normalization schedules.

This paper studies that separation directly. We begin with a deterministic semantic system

$$\mathfrak{S} = (M, \Sigma, \{U_a\}_{a \in \Sigma}, m_0)$$

and an exact physical policy domain C equipped with a projection

$$\Theta : C \rightarrow M, \quad \Theta \delta_a = U_a \Theta.$$

A chosen section

$$\sigma : M \rightarrow C, \quad \Theta \sigma = \text{id}_M,$$

selects one canonical physical representative of each semantic state. The composite

$$N = \sigma \Theta$$

is then a canonical retraction. This elementary construction is the organizing device for the whole theory. It separates the semantic state of a configuration from its position inside the corresponding physical fiber, and it gives a coordinate-free, section-relative definition of representation debt:

$$c \text{ carries debt} \iff c \neq N(c).$$

No additive defect vector, Parikh difference, or stack-length discrepancy is assumed at the abstract level. Such quantities, when available, are coordinate charts on a more general fiber phenomenon.

The next question is local realizability. Even when the semantic update U_a is deterministic and the canonical target $\sigma(U_a(m))$ is unique, a real-time pushdown transition may be unable to reach that target from $\sigma(m)$ in one input-consuming step. The cause is LIFO locality: a bounded observation can inspect and rewrite only a bounded prefix of the physical representation, while an unobserved tail must survive according to the policy. This obstruction is neither semantic ambiguity nor nondeterminism. It is a compatibility problem between a canonical section and a local physical action.

A principal goal of the paper is to classify this failure without tying the theory to one counting language or one literal stack encoding. For a finite-observation tail-injective policy, a semantic target with too many one-step semantic predecessors cannot have all of those predecessors mapped canonically to it: local observations do not have enough collapse capacity. If that coarse capacity test passes, a chosen section may still place two same-target sources in the same local observation class. Finally, even when those sources are locally distinguishable, the physical rewrite of the hidden tail may be incompatible with the chosen canonical target. These give three distinct diagnostic stages:

$$\text{capacity overflow} \longrightarrow \text{observation collision} \longrightarrow \text{tail/action misalignment}.$$

For a standard one-top pushdown rule

$$(q, XT) \xrightarrow{a} (q', \gamma T),$$

the final stage is exact: the untouched tail T must be a suffix of the desired canonical target stack.

The distinction matters because the three stages have different design meaning. Capacity overflow is a category-level impossibility inside the stated bounded-local tail-injective policy class. Observation collision is relative to the chosen allocation of canonical representatives to local observations and may

disappear after re-coordination. Tail misalignment is a finer representation/action incompatibility: semantic predecessor fibers can even be singletons while the chosen stack words remain impossible to connect by the local rewrite.

The same separation then supports the remaining layers of the theory: behavioral safety and containment, causal normalization schedules, section synthesis, and effective verification. In particular, noncanonicity is kept distinct from behavioral failure, and synthesis is formulated through rooted path-outcome coherence rather than through a preferred stack syntax.

1.1 Contributions

The paper is organized around eight theorem families.

- (i) **Semantic fibers and canonical retraction.** Exact lifts and sections induce an idempotent canonical retraction whose fibers are exactly the semantic fibers. Canonical normalization commutes with every semantic word action after retraction.
- (ii) **Bounded-local realizability and obstruction decomposition.** We derive finite-observation collapse-capacity bounds, an exact fixed-section compatibility criterion, and the three-stage distinction between capacity overflow, observation collision, and tail/action misalignment.
- (iii) **Behavioral hull and debt dichotomy.** For a fixed policy we isolate the largest set of physical configurations whose future acceptance behavior agrees with the semantic system. This hull is forward invariant, so genuinely harmless debt cannot later become fatal.
- (iv) **Semantic containment cores.** Forward-invariant semantic traps whose entire physical fibers are terminally correct provide structural certificates of harmless debt. Greatest-fixed-point versions separate ambient full-fiber certification from reachable-fiber certification.
- (v) **Safe reset, fatal frontier, and causal certification.** Local eager normalization choices are related to global correctness through reachable reset domains and finite diagnostic continuations.
- (vi) **Schedule conflict structure.** Under robust fatal certificates, finite optional normalization schedules form an independence complex determined by minimal causal conflicts. Exact weighted-balance families provide nontrivial parameterized examples.
- (vii) **Section synthesis and forced debt.** Transition-closed rooted sections are characterized by rooted path-outcome coherence. This yields finite propagation algorithms, symbolic integer-weight synthesis, and quantitative lower bounds on unavoidable debt at locality-limited boundaries.
- (viii) **Effective verification hierarchy.** We distinguish lightweight structural saturation, exact regular immediate-disagreement saturation, and the general deterministic pushdown equivalence fallback.

The main methodological claim is not that every deterministic pushdown automaton has a distinguished canonical stack representation. Rather, once a semantic quotient, a physical policy category, and a section are made explicit, questions about “forced” noncanonicity, locality, minimality, scheduling, and verification become mathematically well posed.

1.2 Scope and novelty boundary

The word *general* refers to the phenomenon and the policy class, not to all context-free languages or all pushdown automata. No theorem below claims that every correct DPDA admits a chosen global section, that every canonical section is reachable, that every locality obstruction is behaviorally fatal, or that all schedule families are downward closed. Physical minimality is always relative to a stated representation/morphism category.

Several algorithmic ingredients used later are classical, including pushdown reachability saturation and deterministic pushdown language equivalence. The contribution here is the structural reduction that identifies which semantic/physical objects those tools should be applied to. The classical algorithmic results invoked below are cited where used; the mathematical contribution claimed here is the semantic/physical reduction framework rather than those underlying decision procedures.

1.3 Relation to prior work

The semantic/physical separation used here sits near several established lines of automata theory, but it is not identical to any of them. Visibly pushdown languages provide a particularly successful structural restriction on pushdown behavior, and congruence-based work on VPL develops canonical and minimized deterministic representations for important visible subclasses [1, 2, 3]. Our semantic state space M , by contrast, need not be finite, visibly generated, or language-minimal. A section $\sigma : M \rightarrow C$ is not asserted to be a minimal automaton construction; it is a chosen physical representative of each semantic state inside a stated policy category. The main questions are whether that representative is locally realizable, whether noncanonical representatives are forced, and whether such debt matters behaviorally.

There is also a substantial literature on semantic comparison between pushdown processes and finite-state specifications. In particular, Kučera and Mayr give a generic framework for checking full semantic equivalence and for deciding whether a finite-state process is an appropriate quotient of a pushdown process for broad classes of process equivalences [4]. The quotient used in the present paper is instead an explicit deterministic semantic factor supplied as part of the model:

$$\Theta\delta_a = U_a\Theta.$$

We do not claim a new general quotient construction for pushdown processes. Our contribution begins after such a factor and a physical policy category have been fixed, and studies exact sections, locality, debt, scheduling, and synthesis relative to them.

Reachable state-stack configurations have likewise been studied through store languages; in particular, store languages of ordinary one-way pushdown acceptors are known to be regular [5]. That theory concerns which store configurations can occur in accepting computations and the language-theoretic complexity of those configuration sets. Our reachable/ambient distinction uses the same basic configuration objects but asks a different question: how physical configurations lie in semantic fibers and whether a chosen canonical representative can be maintained by bounded-local updates.

The word “reset” in the scheduling part should also not be confused with synchronizing words. Synchronization for deterministic visibly pushdown automata, and for more general deterministic pushdown subclasses, asks for a single input word that brings several starting states to a common state, with various stack constraints [6, 7]. A safe-reset domain here is instead a proof certificate for local normalization events along one run. Similarly, reversible pushdown automata impose backward determinism on whole configurations [8]; the tail-injectivity assumption used in our local collapse-capacity theorem is only a local restriction on how an unobserved tail may be transformed by one bounded observation rule.

The integer-weight synthesis examples are also situated inside the classical theory of one-counter automata. Deterministic one-counter automata and their equivalence problem have been studied since Valiant and Paterson [9]. Accordingly, we do not present counter computation, quotient/remainder encoding, or deterministic one-counter recognition as new in themselves. The result proved here is a representation theorem inside the present exact-lift formalism: for integer translation semantics, a finite modular control skeleton plus a unary quotient stack yields a transition-closed section and therefore a zero-debt physical realization in the stated template.

A terminology distinction is also useful. “Path coherence” is an established term in the Wheeler-graph literature for the convexity of sets of states reached by equally labeled paths [10]. Our synthesis criterion concerns a different property: all semantic paths from a fixed root to the same semantic state must have the same *physical endpoint*. We therefore call it *rooted path-outcome coherence*.

These comparisons motivate a deliberately cautious novelty claim. Several statements in the paper are intentionally organizational rather than priority claims. In particular, the retraction identities following from $\Theta\sigma = \text{id}$, the greatest-fixed-point presentation of a safety hull, and the well-definedness criterion behind rooted path-outcome coherence are elementary once the framework is fixed. Likewise, “independence complex” is used as combinatorial packaging rather than as a new combinatorial object.

The contributions most specific to the present framework are the results that depend essentially on the semantic-section/local-policy interface: finite-observation collapse capacity, the three-stage locality decomposition, the distinction between representation debt and behavioral fatality, the safe-reset/fatal-frontier causal calculus, and the quantitative forced-debt bounds. The literature reviewed above contains close neighboring ideas but no direct formulation of the finite-observation collapse bound in terms of transition-closed semantic sections and tail-injective hidden updates. We therefore use the cautious formulation “we are not aware” and make no absolute priority claim. The verification algorithms used in the last part remain classical pushdown saturation and deterministic pushdown equivalence [11, 12]; the contribution claimed there is the reduction to the appropriate structural or behavioral target.

Notation convention. Throughout the abstract theory, C denotes the physical configuration domain. In the centered weighted-balance case studies the center control is written $\mathbf{C} = D_0$ so that it cannot be confused with that domain or with the optional site labeled C . The symbols $A, \dots, F$ are used only as names of optional sites. For multi-type Dyck semantics, $\mathcal{I}_k = \{1, \dots, k\}$ denotes the finite alphabet of bracket types.

2 Semantic systems, exact lifts, and canonical sections

Definition 2.1 (Deterministic semantic system). A deterministic semantic system is a tuple

$$\mathfrak{S} = (M, \Sigma, \{U_a\}_{a \in \Sigma}, m_0),$$

where M is a semantic state space and every input symbol acts by a total map $U_a : M \rightarrow M$. For a word $w = a_1 \cdots a_n$ we write

$$U_w = U_{a_n} \circ \cdots \circ U_{a_1}, \quad U_\varepsilon = \text{id}_M.$$

The theory imposes no additive, commutative, or counting structure on M . Examples used later include an integer translation system, a finite two-state quotient, and the noncommutative Dyck semantic space $\mathcal{I}_k^* \cup \{\perp\}$.

Definition 2.2 (Exact physical lift). Let C be a physical policy/configuration domain with deterministic input-consuming transitions

$$\delta_a : C \rightarrow C.$$

An exact lift of $\mathfrak{S}$ is a map

$$\Theta : C \rightarrow M$$

such that

$$\boxed{\Theta \delta_a = U_a \Theta} \quad (a \in \Sigma).$$

For a distinguished physical initial state c_0 , let C^{reach} denote the configurations reachable from c_0 by ordinary input words.

Remark 2.3 (Partial machine presentations). The use of total maps $\delta_a : C \rightarrow C$ is a semantic convenience, not a restriction to syntactically total PDA tables. A partial deterministic presentation can be totalized while preserving exactness by adjoining a tagged sink copy $\dagger_m$ of each semantic state m : put

$$\Theta(\dagger_m) = m, \quad \delta_a(\dagger_m) = \dagger_{U_a(m)},$$

and send an otherwise undefined move from c on input a to $\dagger_{U_a(\Theta(c))}$. Terminal observation may reject all tagged sink configurations. Thus the results below can be read either for an intrinsically total policy or for this exact totalization.

The distinction between C and C^{reach} is essential. The policy domain may contain syntactically meaningful canonical representatives that are not reached by the selected baseline policy. Treating all canonical points as reachable would make several otherwise natural equivalences false.

Definition 2.4 (Canonical section). A canonical section is a map

$$\sigma : M \rightarrow C$$

satisfying

$$\boxed{\Theta \sigma = \text{id}_M}.$$

Its image $\text{im} \sigma$ is the chosen canonical physical representation. A section point is *internal* to the current policy when it belongs to C^{reach} and *external* otherwise.

Definition 2.5 (Canonical retraction and representation debt). Define

$$\boxed{N = \sigma \Theta : C \rightarrow C}.$$

A configuration c is *debt-free* when $c = N(c)$ and carries *representation debt* otherwise.

This definition is intentionally coordinate free. In a literal weighted balance representation, debt may later be charted by quantity, order, or control defects. In a Dyck lift it may be an arbitrary residual stack word inside an absorbing dead fiber. The abstract theory identifies all of these as the same phenomenon: noncanonical position inside one semantic fiber.

2.1 Main Theorem I: semantic fibers and canonical retraction

Theorem 2.6 (Canonical fiber and retraction). *For every exact lift with section:*

$$N^2 = N, \tag{1}$$

$$\Theta N = \Theta, \tag{2}$$

$$\text{Fix}(N) = \text{im}\sigma. \tag{3}$$

Moreover, for all $c, c' \in C$,

$$\boxed{\Theta(c) = \Theta(c') \iff N(c) = N(c').}$$

Hence the fibers of N are exactly the semantic fibers $\Theta^{-1}(m)$, each with distinguished basepoint $\sigma(m)$.

Proof. Since $\Theta\sigma = \text{id}_M$,

$$N^2 = \sigma\Theta\sigma\Theta = \sigma\Theta = N,$$

and

$$\Theta N = \Theta\sigma\Theta = \Theta.$$

If $c = \sigma(m)$ then

$$N(c) = \sigma\Theta\sigma(m) = \sigma(m) = c,$$

so $\text{im}\sigma \subseteq \text{Fix}(N)$. Conversely, if $N(c) = c$, then $c = \sigma(\Theta(c)) \in \text{im}\sigma$.

If $\Theta(c) = \Theta(c')$, applying σ gives $N(c) = N(c')$. Conversely, if $N(c) = N(c')$, applying Θ and using $\Theta N = \Theta$ yields $\Theta(c) = \Theta(c')$. $\square$

Theorem 2.7 (Canonical absorption). *For every word $w \in \Sigma^*$,*

$$\boxed{N\delta_w = N\delta_w N = \sigma U_w \Theta.}$$

Proof. Exactness extends inductively to words:

$$\Theta\delta_w = U_w \Theta.$$

Therefore

$$N\delta_w = \sigma\Theta\delta_w = \sigma U_w \Theta.$$

Also

$$N\delta_w N = \sigma U_w \Theta N = \sigma U_w \Theta,$$

because $\Theta N = \Theta$. $\square$

Remark 2.8. Theorem 2.7 is not a Church–Rosser confluence statement. Two physical histories may be entirely different. The theorem says only that after canonical retraction they have the uniquely determined semantic endpoint.

Theorem 2.9 (Zero-debt/raw-conjugacy equivalence). *Assume:*

- (a) $c_0 = \sigma(m_0)$;
- (b) every state of the semantic component under discussion is reachable from m_0 ;
- (c) C^{reach} is the ordinary-input reachable set from c_0 .

Then

$$\boxed{\delta_a \sigma = \sigma U_a \quad \forall a \in \Sigma}$$

if and only if

$$\boxed{C^{\text{reach}} = \sigma(M)}.$$

Proof. Assume first that $\delta_a \sigma = \sigma U_a$ for every input symbol. By induction,

$$\delta_w \sigma(m_0) = \sigma U_w(m_0).$$

Hence every ordinary-input reachable configuration lies in $\sigma(M)$. Conversely, semantic reachability gives, for each $m \in M$, a word w with $U_w(m_0) = m$, so $\sigma(m) = \delta_w(c_0)$ is reachable. Thus $C^{\text{reach}} = \sigma(M)$.

Conversely suppose $C^{\text{reach}} = \sigma(M)$. For $m \in M$ and $a \in \Sigma$, the configuration $\delta_a \sigma(m)$ is reachable and therefore equals $\sigma(m')$ for some semantic state m' . Exactness gives

$$m' = \Theta \delta_a \sigma(m) = U_a(m),$$

hence $\delta_a \sigma(m) = \sigma U_a(m)$. □

Remark 2.10 (Why mere reachable bijectivity is insufficient). A restriction $\Theta|_{C^{\text{reach}}}$ can be bijective even when the preselected section uses external representatives. The converse direction in Theorem 2.9 therefore requires equality $C^{\text{reach}} = \sigma(M)$, not merely the existence of some inverse to $\Theta|_{C^{\text{reach}}}$.

2.2 Normalization inside a fiber

Definition 2.11 (Semantic-preserving normalization). A physical normalization map $\lambda : C \rightarrow C$ is semantic preserving when

$$\Theta \lambda = \Theta.$$

Then automatically

$$\boxed{N \lambda = N}.$$

Thus normalization may alter control state, stack order, stack length, or any other physical coordinate, but it does not alter the semantic point. The canonical retraction N is the distinguished endpoint map induced by the chosen section; other administrative normalizers need not equal N on every intermediate configuration.

3 Bounded-local LIFO realizability

The semantic target $\sigma(U_a(m))$ is unique once the section is fixed. The physical question is whether a legal local transition can reach it directly from $\sigma(m)$. We formulate this first for ordinary one-top pushdown rewriting and then for a more general finite-observation tail-injective policy.

3.1 One-top rewriting

Definition 3.1 (One-top local policy). A one-top configuration is written

$$(q, XT),$$

where q is finite control, X is the visible top stack symbol, and T is the untouched tail. On input a , a deterministic one-top rule has the form

$$\boxed{(q, XT) \xrightarrow{a} (q', \gamma T)}.$$

Lemma 3.2 (Uniform one-top portability). *Suppose a family of canonical source configurations exposes the same local observation*

$$(q, a, X)$$

and has the form (q, XT_i) . Let the desired canonical targets be (q_i^, S_i^*) . One deterministic one-top rule realizes all of these targets if and only if there exist one control state q' and one word γ such that*

$$q_i^* = q' \quad \text{and} \quad \boxed{S_i^* = \gamma T_i} \quad \text{for every } i.$$

Proof. A deterministic transition cell indexed by (q, a, X) has one target control state and one replacement word. Therefore every successor has the form $(q', \gamma T_i)$, proving necessity. The same rule proves sufficiency. $\square$

Corollary 3.3 (Suffix criterion). *For a single one-top source (q, XT) , a desired target stack S^* is directly realizable in one transition if and only if*

$$\boxed{T \text{ is a suffix of } S^*}.$$

The lemma identifies a local physical fact that is independent of weighted counting. It will reappear below as the one-top specialization of the general tail-alignment condition.

3.2 Finite-observation tail-injective policies

Definition 3.4 (Finite-observation tail-injective policy). Fix an input symbol a . Assume every canonical source configuration can be decomposed as

$$c = (o, T),$$

where o lies in a finite set of local observations Ω_a and T is the hidden tail. The physical update associated with o has the form

$$\boxed{(o, T) \mapsto (h_a(o), F_{a,o}(T))},$$

where $F_{a,o}$ is injective.

The injectivity hypothesis captures the basic locality restriction that one local rule cannot erase arbitrary distinctions in an unobserved tail. For a standard one-top rule, o contains the visible control/top information and $F_{a,o}$ is simply the identity on the untouched suffix.

Theorem 3.5 (Finite-observation collapse capacity). *Assume the section is transition closed on input a :*

$$\delta_a \sigma = \sigma U_a.$$

Then for every semantic target m^ ,*

$$\boxed{|U_a^{-1}(m^*)| \leq |\Omega_a|}.$$

Proof. Let $m_1, m_2 \in U_a^{-1}(m^*)$ have canonical sources with the same local observation o :

$$\sigma(m_i) = (o, T_i).$$

Transition closure gives

$$\delta_a \sigma(m_1) = \sigma(m^*) = \delta_a \sigma(m_2).$$

Hence

$$F_{a,o}(T_1) = F_{a,o}(T_2).$$

Injectivity yields $T_1 = T_2$, so $\sigma(m_1) = \sigma(m_2)$ and therefore $m_1 = m_2$. Thus distinct semantic predecessors of m^* must occupy distinct observations, giving the bound. $\square$

Corollary 3.6 (Infinite-collapse obstruction). *If some one-symbol semantic predecessor fiber $U_a^{-1}(m^*)$ is infinite, then no transition-closed section exists in any fixed finite-observation tail-injective policy category.*

Corollary 3.7 (Bounded- d -top pushdown capacity). *If the physical policy uses finite control Q , stack alphabet Γ , and observes at most the first d stack symbols, then*

$$|U_a^{-1}(m^*)| \leq |Q| \sum_{j=1}^d |\Gamma|^j$$

is necessary for a transition-closed section.

3.3 Main Theorem II: obstruction decomposition

For a fixed section, define the observation exposed by its canonical source:

$$\text{obs}_a(m) \in \Omega_a.$$

Write

$$\sigma(m) = (o_m, T_m).$$

Theorem 3.8 (Three-stage bounded-local obstruction decomposition). *Fix an input symbol a , a finite-observation tail-injective physical policy, and a chosen section σ .*

The direct-canonical realizability problem admits the following ordered diagnosis.

Stage I: Semantic capacity overflow. *If some semantic target m^* satisfies*

$$|U_a^{-1}(m^*)| > |\Omega_a|,$$

then no transition-closed section can exist in the stated policy category.

Stage II: Chosen-section observation collision. *Assume the category is not excluded by Stage I. If distinct semantic sources $m_1 \neq m_2$ satisfy*

$$U_a(m_1) = U_a(m_2)$$

and

$$\text{obs}_a(m_1) = \text{obs}_a(m_2),$$

then the chosen section cannot be transition closed.

Stage III: Local alignment failure. *Assume the chosen section is observation-injective on every same-target semantic predecessor fiber. Then the remaining condition for direct canonical realization is*

$$\sigma(U_a(m)) = (h_a(o_m), F_{a,o_m}(T_m))$$

for every semantic source m . Failure of this equation at any source is a tail/action alignment obstruction.

Consequently, for a fixed section and policy, transition closure on input a is equivalent to absence of Stage-II collisions together with satisfaction of every Stage-III alignment equation.

Proof. Stage I is Theorem 3.5.

For Stage II, suppose

$$\sigma(m_i) = (o, T_i)$$

with one common observation o , and suppose both semantic successors equal m^* . If the section were transition closed, both physical outputs would equal $\sigma(m^*)$. Injectivity of $F_{a,o}$ would then force $T_1 = T_2$, hence $\sigma(m_1) = \sigma(m_2)$ and therefore $m_1 = m_2$, a contradiction.

For Stage III, once the source observation has been fixed, the deterministic physical successor is exactly

$$(h_a(o_m), F_{a,o_m}(T_m)).$$

Thus it is canonical exactly when it equals the chosen canonical target $\sigma(U_a(m))$. If this equality holds for every source, then $\delta_a\sigma = \sigma U_a$ by definition. Conversely, transition closure implies every such equality. $\square$

Corollary 3.9 (One-top specialization). *For a standard one-top rule*

$$(q, XT) \xrightarrow{a} (q', \gamma T),$$

Stage III of Theorem 3.8 is precisely the uniform suffix/tail criterion of Lemma 3.2.

Theorem 3.10 (Locality–debt–recoordination trilemma). *If direct canonical realization fails for a fixed semantic system, fixed bounded-local physical policy, and fixed section, then exact semantics cannot retain simultaneously:*

- (i) the same local observation regime,
- (ii) the same canonical section,
- (iii) direct canonical successor behavior.

At least one response is necessary:

refine physical observation/information

or

allow representation debt

or

re-coordinate the canonical representation.

Proof. The failure hypothesis says that the current local transition cannot map the current canonical source to the current canonical semantic target. Preserving all three listed choices would require exactly that impossible transition. Therefore at least one choice must change. $\square$

3.4 An obstruction atlas

The three diagnostic stages are logically distinct. Four small examples realize the four qualitatively different outcomes.

Example 3.11 (One-type Dyck: all tests pass). Let Dyck_1 use semantic states $\mathbb{N} \cup \{\perp\}$, with an opening incrementing the live height, a matching close decrementing positive height, and underflow entering the absorbing dead state $\perp$. A natural physical lift distinguishes the live-bottom observation from dead control. The only nontrivial close-collapse into $\perp$ has two semantic sources when the dead state itself is included, but the canonical representatives expose distinct local observations. The natural section is transition closed. This is the regime

$$\boxed{\text{capacity PASS, allocation PASS, alignment PASS.}}$$

Example 3.12 (A pure observation-allocation obstruction). Consider the regular language 1^* with semantic states $\mathbf{L}$ (live) and $\mathbf{D}$ (absorbing dead):

$$U_0(\mathbf{L}) = U_0(\mathbf{D}) = \mathbf{D}, \quad U_1(\mathbf{L}) = \mathbf{L}, \quad U_1(\mathbf{D}) = \mathbf{D}.$$

Use one exact one-top physical lift on

$$C = \{AZ, ABZ, BZ, Z\}$$

with

$$AZ \xrightarrow{0} Z, \quad ABZ \xrightarrow{0} BZ,$$

and all B -top and bottom configurations stable in the dead fiber; input 1 preserves each listed configuration. Define

$$\Theta(AZ) = \mathbf{L}, \quad \Theta(ABZ) = \Theta(BZ) = \Theta(Z) = \mathbf{D}.$$

The section

$$\sigma_{\text{bad}}(\mathbf{L}) = AZ, \quad \sigma_{\text{bad}}(\mathbf{D}) = ABZ$$

places the two 0-predecessors of $\mathbf{D}$ under the same visible top A . Their hidden tails Z and BZ cannot both be mapped by the same tail-injective pop to the single canonical target ABZ . Stage I passes but Stage II fails.

In the *same exact physical lift*, the re-sectioning

$$\sigma_{\text{good}}(\mathbf{L}) = AZ, \quad \sigma_{\text{good}}(\mathbf{D}) = Z$$

is transition closed. Thus Stage II is genuinely section/allocation relative.

Example 3.13 (Pure tail/alignment obstruction). Consider the weighted language

$$L_{2,3,-5} = \{w \in \{a, b, c\}^* : 2N_a(w) + 3N_b(w) - 5N_c(w) = 0\},$$

where $N_x(w)$ counts occurrences of the symbol x in w . with semantic residual system $M = \mathbb{Z}$ and translations

$$U_a(r) = r + 2, \quad U_b(r) = r + 3, \quad U_c(r) = r - 5.$$

Every one-symbol semantic map is bijective, so every predecessor fiber has size one. Stages I and II are therefore vacuous.

For a shortest-completion block-literal section, however, residual -5 has one a -demand and one b -demand. Any block ordering places either b before a or a before b . In the first case, reading a requires a canonical target containing only b while the hidden a tail must survive; in the second case, reading b gives the symmetric contradiction. Thus Stage III fails despite singleton semantic fibers. A quotient/remainder one-counter re-coordination removes this obstruction.

Example 3.14 (Multi-type Dyck: category-level capacity failure). Let

$$M_k = \mathcal{I}_k^* \cup \{\perp\}$$

be the multi-type Dyck semantic core, with top at the left; we write Dyck_k for this semantic system. Opening type i maps s to is . A close c_j removes a matching top j and sends underflow or a mismatching top to $\perp$. Fix $i \neq j$. Then

$$U_{c_j}(it) = \perp \quad \text{for every } t \in \mathcal{I}_k^*.$$

Thus

$$|U_{c_j}^{-1}(\perp)| = \infty.$$

By Theorem 3.5, no fixed finite-observation tail-injective bounded-local policy can possess a globally transition-closed section. This is a Stage-I category-level obstruction, not a defect of one particular block ordering.

The four examples give the table

example	capacity	allocation	alignment
Dyck_1	PASS	PASS	PASS
$1^*, \sigma_{\text{bad}}$	PASS	FAIL	--
$L_{2,3,-5}$ block section	PASS	PASS	FAIL
$\text{Dyck}_k, k \geq 2$	FAIL	--	--

and show that collapse capacity is a coarse necessary screen rather than a complete local realizability criterion.

4 Behavioral correctness and representation debt

The preceding sections separate semantic state from physical representation and classify local failures of direct canonical realization. A different question now becomes unavoidable: when a physical configuration is noncanonical, is that noncanonicity behaviorally significant?

The answer is no in general. A configuration may carry representation debt while still inducing exactly the same future acceptance behavior as its semantic state. Conversely, a noncanonical configuration may already be doomed: the error is not yet visible at the current endpoint, but some finite continuation will expose it. This section makes that distinction exact.

Throughout this section a physical policy is fixed. We suppress the policy index unless several policies are compared.

Definition 4.1 (Terminal observables). Let

$$f : M \rightarrow O$$

be the semantic terminal observable and

$$\alpha : C \rightarrow O$$

the physical terminal observable, where O is any finite set of terminal outcomes. In a recognizer one may take $O = \{0, 1\}$.

Definition 4.2 (Immediate terminal disagreement). The immediate disagreement set is

$$\mathcal{D} = \{c \in C : \alpha(c) \neq f(\Theta(c))\}.$$

A configuration in $\mathcal{D}$ is already wrong if the computation is terminated immediately.

Definition 4.3 (Behavioral safe hull). The behavioral safe hull is

$$\mathcal{B} = \{c \in C : \alpha(\delta_w(c)) = f(U_w(\Theta(c))) \text{ for every } w \in \Sigma^*\}.$$

Thus $c \in \mathcal{B}$ exactly when the physical and semantic systems have the same terminal behavior after every continuation from c .

The behavioral hull is the exact object needed to separate harmless representation debt from latent recognition failure. In the fixed-point formulas below, $\nu X.\Phi(X)$ denotes the greatest fixed point of the monotone operator Φ .

4.1 Main Theorem III: behavioral hull and debt dichotomy

Proposition 4.4 (Terminal factorization). *The identity*

$$\alpha = f\Theta$$

holds on all of C if and only if:

- (a) α is constant on every semantic fiber of Θ ; and
- (b)

$$\alpha\sigma = f.$$

Proof. If $\alpha = f\Theta$, then equal semantic projection implies equal physical terminal value, and

$$\alpha\sigma = f\Theta\sigma = f.$$

Conversely, let $c \in C$. Fiber constancy gives

$$\alpha(c) = \alpha(\sigma(\Theta(c))).$$

Using $\alpha\sigma = f$ yields

$$\alpha(c) = f(\Theta(c)).$$

□

Theorem 4.5 (Behavioral hull characterization). *Define the monotone operator*

$$\Phi(X) = \{c \in C : \alpha(c) = f(\Theta(c)) \text{ and } \delta_a(c) \in X \text{ for all } a \in \Sigma\}.$$

Then

$$\mathcal{B} = \nu X.\Phi(X).$$

Equivalently,

$$\mathcal{B} = C \setminus \text{Pre}^*(\mathcal{D}),$$

where $\text{Pre}^(\mathcal{D})$ is the set of configurations from which some finite input word reaches $\mathcal{D}$.*

Proof. Suppose first that $c \in \mathcal{B}$. Taking the empty continuation gives

$$\alpha(c) = f(\Theta(c)).$$

For every $a \in \Sigma$ and continuation z ,

$$\alpha(\delta_z(\delta_a(c))) = \alpha(\delta_{az}(c)) = f(U_{az}(\Theta(c))).$$

Exactness gives

$$U_{az}(\Theta(c)) = U_z(\Theta(\delta_a(c))),$$

so $\delta_a(c) \in \mathcal{B}$. Hence

$$\mathcal{B} \subseteq \Phi(\mathcal{B}).$$

Conversely, let X satisfy $X \subseteq \Phi(X)$. Starting from any $c \in X$, induction on word length shows that every descendant $\delta_w(c)$ remains in X and is therefore immediately terminally correct. Hence

$$\alpha(\delta_w(c)) = f(\Theta(\delta_w(c))) = f(U_w(\Theta(c)))$$

for every word w . Thus $X \subseteq \mathcal{B}$. Taking $X = \Phi(\mathcal{B})$ is legitimate because $\mathcal{B} \subseteq \Phi(\mathcal{B})$ was proved above; the same argument gives

$$\Phi(\mathcal{B}) \subseteq \mathcal{B}.$$

Hence $\Phi(\mathcal{B}) = \mathcal{B}$. Since every post-fixed set $X \subseteq \Phi(X)$ is contained in $\mathcal{B}$, the set $\mathcal{B}$ is the greatest fixed point.

For the predecessor formulation, $c \notin \mathcal{B}$ exactly when there exists a word w for which

$$\alpha(\delta_w(c)) \neq f(U_w(\Theta(c))).$$

By exactness the right side is

$$f(\Theta(\delta_w(c))),$$

so $\delta_w(c) \in \mathcal{D}$. This is precisely $c \in \text{Pre}^*(\mathcal{D})$. □

Corollary 4.6 (Root correctness). *For a fixed policy with physical root c_0 and semantic root $m_0 = \Theta(c_0)$,*

$$\boxed{\text{the policy is behaviorally correct from the root} \iff c_0 \in \mathcal{B}.}$$

Theorem 4.7 (Forward invariance of behavioral safety). *For every word w ,*

$$\boxed{c \in \mathcal{B} \implies \delta_w(c) \in \mathcal{B}.}$$

Proof. The one-symbol case was established in the proof of Theorem 4.5. Iterating it over the letters of w gives the claim. □

Theorem 4.8 (Debt birth dichotomy). *Let c be noncanonical:*

$$c \neq N(c).$$

Exactly one of the following holds.

(i)

$$c \in \mathcal{B}.$$

Then every descendant $\delta_w(c)$ remains in $\mathcal{B}$; the debt is permanently behaviorally harmless under the fixed policy.

(ii)

$$c \notin \mathcal{B}.$$

Then there exists a finite word z such that

$$\boxed{\alpha(\delta_z(c)) \neq f(U_z(\Theta(c)))}.$$

The configuration is already behaviorally unsafe when created; the suffix z merely exposes that pre-existing unsafety.

Proof. The alternatives $c \in \mathcal{B}$ and $c \notin \mathcal{B}$ are exhaustive and disjoint. The first conclusion follows from Theorem 4.7. In the second case, Theorem 4.5 gives

$$c \in \text{Pre}^*(\mathcal{D}),$$

so some finite continuation reaches immediate disagreement. $\square$

Remark 4.9 (No safe-to-fatal escape). Theorem 4.8 rules out a tempting but incorrect causal picture. A debt configuration that is genuinely behaviorally safe does not later “escape” from safety and become fatal under the same policy. What may happen is that a configuration is born outside $\mathcal{B}$ while the finite continuation that reveals the error is read only later.

4.2 Two opposite debt regimes

Example 4.10 (Multi-type Dyck: forced but harmless debt). Let

$$M_k = \mathcal{I}_k^* \cup \{\perp\}$$

be the multi-type Dyck semantic system from the obstruction atlas. The natural exact physical lift mirrors a live semantic stack word in a good control state. On a mismatch or underflow it enters a dead control state, and after a mismatch it may leave the old physical stack in place.

Choose every dead-control configuration to reject, and let the semantic terminal observable reject $\perp$. Since

$$U_a(\perp) = \perp \quad \text{for all } a,$$

every physical configuration in the dead semantic fiber has the correct future terminal behavior. Thus mismatch debt can be physically large and unavoidable while remaining behaviorally harmless.

Example 4.11 (Centered weighted balance: latent-fatal debt). In the centered 2:1 schedule family, writing $\mathbf{C}$ for the center control, one eager interaction reaches the noncanonical configuration

$$c = (\mathbf{C}, 1110)$$

while the semantic residual is the same as that of the canonical stack 1. Appending one input symbol 1 yields a residual-zero nonaccepting physical configuration. Hence

$$c \notin \mathcal{B}.$$

The final 1 does not make the debt unsafe; it is a finite diagnostic continuation witnessing that the debt was already outside the behavioral hull when created.

The analogous exact 3:2, 5:2, and odd- p :2 schedule witnesses have the same logical form.

These examples establish an important separation:

$$\boxed{\text{representation debt} \neq \text{behavioral failure}}.$$

Locality may force debt even when every future answer is correct, while a different debt state may already encode a latent counterexample.

5 Semantic containment cores

The behavioral hull is exact, but it is a physical object: deciding membership may require reasoning about all physical continuations from a specific configuration. A coarser semantic certificate is often useful. The idea is to identify semantic regions whose entire physical preimages are terminally correct and cannot semantically escape that correctness.

5.1 Main Theorem IV: semantic containment

Theorem 5.1 (Semantic trap containment). *Let $S \subseteq M$ satisfy*

$$U_a(S) \subseteq S \quad \text{for every } a \in \Sigma.$$

Assume also that terminal factorization holds on the entire physical preimage:

$$\alpha(c) = f(\Theta(c)) \quad \text{for every } c \in \Theta^{-1}(S).$$

Then

$$\boxed{\Theta^{-1}(S) \subseteq \mathcal{B}.}$$

Proof. Take $c \in \Theta^{-1}(S)$ and any word w . Exactness gives

$$\Theta(\delta_w(c)) = U_w(\Theta(c)).$$

Forward invariance of S implies that the right side lies in S . Therefore $\delta_w(c) \in \Theta^{-1}(S)$, where the assumed terminal factorization is valid:

$$\alpha(\delta_w(c)) = f(\Theta(\delta_w(c))) = f(U_w(\Theta(c))).$$

Thus $c \in \mathcal{B}$. □

Corollary 5.2 (Absorbing-fiber harmlessness). *Let $b \in M$ be absorbing:*

$$U_a(b) = b \quad \forall a \in \Sigma.$$

If every configuration in $\Theta^{-1}(b)$ has the correct terminal value $f(b)$, then

$$\boxed{\Theta^{-1}(b) \subseteq \mathcal{B}.}$$

Example 4.10 is exactly this situation with $b = \perp$.

Definition 5.3 (Full-fiber terminally correct states). Define

$$\boxed{G_0 = \{m \in M : \alpha(c) = f(m) \text{ for every } c \in \Theta^{-1}(m)\}.$$

Define the monotone operator

$$\boxed{\Psi(S) = G_0 \cap \bigcap_{a \in \Sigma} U_a^{-1}(S).$$

Theorem 5.4 (Maximal fiberwise semantic containment core). *Let*

$$\boxed{\mathcal{T} = \nu S. \Psi(S).$$

Then:

(a)

$$\mathcal{T} = \{m \in M : U_w(m) \in G_0 \text{ for every } w \in \Sigma^*\};$$

(b) $\mathcal{T}$ is the greatest forward-invariant subset of G_0 ;

(c)

$$\boxed{\Theta^{-1}(\mathcal{T}) \subseteq \mathcal{B}.}$$

Proof. The operator Ψ is the usual safety operator: a state remains in a post-fixed set exactly when it is immediately in G_0 and all one-symbol successors remain in the set. Its greatest fixed point is therefore exactly the set of semantic states whose entire forward orbit remains in G_0 . This proves (a) and (b). Part (c) follows from Theorem 5.1. $\square$

Remark 5.5 (The containment core is deliberately conservative). The inclusion

$$\Theta^{-1}(\mathcal{T}) \subseteq \mathcal{B}$$

need not be equality. A semantic fiber may contain an ambient physical representative that is terminally wrong even though the representative actually reached by the selected policy is behaviorally safe. Because G_0 quantifies over the entire ambient fiber, one unreachable bad representative can remove the whole semantic state from $\mathcal{T}$.

Example 5.6 (Ambient contamination). Return to the language 1^* . Let the reachable physical states be

$$AZ, \quad Z,$$

with projection

$$\Theta(AZ) = L, \quad \Theta(Z) = D,$$

and with the correct terminal values

$$\alpha(AZ) = 1, \quad \alpha(Z) = 0.$$

Add one unreachable ambient state BZ in the dead semantic fiber:

$$\Theta(BZ) = D,$$

but deliberately set

$$\alpha(BZ) = 1.$$

The rooted machine is still correct because BZ is unreachable. However, $D \notin G_0$, and since the live state reaches D on input 0, the full-fiber containment core becomes empty. Thus ambient full-fiber certification can be strictly smaller than actual rooted behavioral correctness.

5.2 Reachable-fiber refinement

Fix the physical root c_0 , and let C^{reach} be the ordinary-input reachable physical set. Put

$$M^{\text{reach}} = \Theta(C^{\text{reach}}).$$

Definition 5.7 (Reachable-fiber containment core). Define

$$G_0^{\text{reach}} = \left\{ m \in M^{\text{reach}} : \alpha(c) = f(m) \text{ for every } c \in C^{\text{reach}} \cap \Theta^{-1}(m) \right\}.$$

On subsets of M^{reach} define

$$\Psi^{\text{reach}}(S) = G_0^{\text{reach}} \cap \bigcap_{a \in \Sigma} U_a^{-1}(S),$$

and let

$$\mathcal{T}^{\text{reach}} = \nu S. \Psi^{\text{reach}}(S).$$

Theorem 5.8 (Reachable semantic containment).

$$C^{\text{reach}} \cap \Theta^{-1}(\mathcal{T}^{\text{reach}}) \subseteq \mathcal{B}.$$

Moreover,

$$\mathcal{T} \cap M^{\text{reach}} \subseteq \mathcal{T}^{\text{reach}}.$$

Proof. Take

$$c \in C^{\text{reach}} \cap \Theta^{-1}(\mathcal{T}^{\text{reach}}).$$

Every descendant of c is still reachable from c_0 . Exactness and forward invariance of $\mathcal{T}^{\text{reach}}$ keep its semantic projection in G_0^{reach} , so every descendant is immediately terminally correct. Therefore $c \in \mathcal{B}$.

For the second claim, full-fiber terminal correctness implies reachable-fiber terminal correctness. Hence every forward-invariant subset of G_0 remains a valid candidate for the reachable operator on the reachable semantic component. Maximality of $\mathcal{T}^{\text{reach}}$ yields the inclusion. $\square$

Theorem 5.9 (Root correctness via reachable containment). *Let $m_0 = \Theta(c_0)$. Then*

$$c_0 \in \mathcal{B} \iff m_0 \in \mathcal{T}^{\text{reach}}.$$

Proof. If $m_0 \in \mathcal{T}^{\text{reach}}$, then Theorem 5.8 gives $c_0 \in \mathcal{B}$.

Conversely, assume $c_0 \in \mathcal{B}$. By Theorem 4.7, every configuration reachable from c_0 also lies in $\mathcal{B}$ and is therefore immediately terminally correct. Thus

$$G_0^{\text{reach}} = M^{\text{reach}}.$$

The reachable semantic component is forward invariant, so its greatest forward-invariant subset is itself:

$$\mathcal{T}^{\text{reach}} = M^{\text{reach}}.$$

In particular $m_0 \in \mathcal{T}^{\text{reach}}$. $\square$

Remark 5.10 (Three correctness scales). The theory now contains three increasingly precise objects:

$$\mathcal{T} \rightsquigarrow \mathcal{T}^{\text{reach}} \rightsquigarrow \mathcal{B}.$$

The first is an ambient full-fiber semantic certificate, the second removes unreachable physical contamination, and the third is the exact physical behavioral object. The first two are useful precisely because they may be easier to certify than full behavioral equivalence.

5.3 Debt taxonomy after behavioral refinement

The locality theory and the behavioral theory classify different aspects of the same physical deviation. It is therefore useful to keep two axes separate.

Representation origin. Debt may be representation-relative and removable by re-coordination, as in the block-literal section for $L_{2,3,-5}$, or it may be forced by a bounded locality category, as in multi-type Dyck.

Behavioral status. Independently, debt may lie in $\mathcal{B}$ and be permanently harmless, or it may lie outside $\mathcal{B}$ and admit a finite diagnostic continuation.

Thus:

$$\boxed{\text{why debt exists} \quad \text{and} \quad \text{whether debt is fatal}}$$

are logically orthogonal questions.

This distinction is essential in the causal scheduling theory. An eager normalization step may create a structurally noncanonical target without causing any language error; schedule failure occurs only when a reachable unsafe activation lands outside the behavioral hull.

6 Causal scheduling

The preceding sections classify configurations. We now classify policies. The motivating situation is a baseline deterministic real-time pushdown normalizer equipped with finitely many optional eager local actions. Each optional action is semantically exact: it changes only the physical representation chosen for the same semantic successor. A *schedule* selects which optional eager actions are enabled.

Let

$$\mathcal{E} = \{e_1, \dots, e_k\}$$

be a finite optional-site universe. For every

$$H \subseteq \mathcal{E}$$

let δ^H be the resulting deterministic physical dynamics. Every schedule is an exact lift of the same semantic dynamics:

$$\Theta \delta_a^H = U_a \Theta \quad (a \in \Sigma).$$

Write α_H for its physical terminal observable (typically the same observable for all schedules), $\mathcal{B}_H$ for its behavioral hull, and

$$\text{Adm}(\mathcal{E}) = \{H \subseteq \mathcal{E} : \text{the policy } H \text{ is correct from the root}\}.$$

The central issue is that direct canonicity and behavioral correctness are different. An eager action may leave a chosen structural reset repertoire without becoming fatal, while a different eager action may land immediately outside the behavioral hull. The safe-reset layer separates these cases.

6.1 Main Theorem V: safe reset and the fatal frontier

Definition 6.1 (Wordwise safe-reset domain). Let

$$\mathfrak{H} \subseteq 2^{\mathcal{E}}$$

be a schedule subfamily. A set

$$\mathcal{R} \subseteq \text{im}\sigma$$

is a *wordwise safe-reset domain* for $\mathfrak{H}$ if the following implication holds for every $H \in \mathfrak{H}$ and every input word w : if every eager target actually produced along the H -run of w lies in $\mathcal{R}$, then the run has the correct semantic terminal result. The case $\mathfrak{H} = 2^{\mathcal{E}}$ is called a safe-reset domain for the full schedule family.

In concrete systems this property is typically proved by decomposing a run into baseline episodes between canonical reset points and inducting over the episodes. The abstract theory uses only the resulting wordwise certificate.

Proposition 6.2 (Safe-reset certificate). *Let $\mathcal{R}$ be a wordwise safe-reset domain for a subfamily $\mathfrak{H}$, and let $H \in \mathfrak{H}$. If every reachable eager application under H lands in $\mathcal{R}$, then*

$$H \in \text{Adm}(\mathcal{E}).$$

Proof. Apply Definition 6.1 to every input word. □

Definition 6.3 (Actual eager targets and structural unsafe frontier). For a schedule H , let

$$\mathcal{A}_H$$

be the set of physical configurations that occur as targets of eager transitions on runs reachable from the root. Define

$$U_{\mathcal{R}}(H) = \mathcal{A}_H \setminus \mathcal{R}.$$

The set $U_{\mathcal{R}}(H)$ is a structural frontier: it records actual eager targets that leave the chosen proof-safe reset domain. Membership in this set alone does not imply behavioral failure.

Definition 6.4 (Reachable behavioral slack). The reachable slack outside the reset certificate is

$$\text{Slack}_{\mathcal{R}}(H) = U_{\mathcal{R}}(H) \cap \mathcal{B}_H.$$

Thus slack consists of actually reached eager targets that are structurally outside $\mathcal{R}$ but behaviorally safe.

Theorem 6.5 (Fatal Frontier). *Assume $\mathcal{R}$ is a wordwise safe-reset domain for the full family $2^{\mathcal{E}}$. Then for every schedule $H \subseteq \mathcal{E}$,*

$$H \notin \text{Adm}(\mathcal{E}) \iff U_{\mathcal{R}}(H) \setminus \mathcal{B}_H \neq \emptyset.$$

Equivalently, every bad schedule has a reachable eager activation that both leaves the structural reset domain and is already behaviorally unsafe.

Proof. Assume first that $H \notin \text{Adm}(\mathcal{E})$ and choose a counterexample word w . If every eager target on the H -run of w lay in $\mathcal{R}$, the wordwise safe-reset property would make the run correct, a contradiction. Hence some eager target leaves $\mathcal{R}$. Let c be the first such target and write

$$w = uz,$$

where u reaches c immediately after that eager step. The remaining suffix z still produces the terminal disagreement of the counterexample. Therefore z distinguishes the physical continuation from c from the semantic continuation determined by $\Theta(c)$, so

$$c \notin \mathcal{B}_H.$$

Thus

$$c \in U_{\mathcal{R}}(H) \setminus \mathcal{B}_H.$$

Conversely, suppose

$$c \in U_{\mathcal{R}}(H) \setminus \mathcal{B}_H.$$

Because c is an actual eager target, some prefix u reaches it under H . Since $c \notin \mathcal{B}_H$, Theorem 4.5 provides a finite continuation z with

$$\alpha_H(\delta_z^H(c)) \neq f(U_z(\Theta(c))).$$

Then uz is a counterexample from the root. Hence $H \notin \text{Adm}(\mathcal{E})$. □

Corollary 6.6 (Activation tightness). *The structural reset certificate is exact on the schedule family if*

$$\text{Slack}_{\mathcal{R}}(H) = \emptyset \quad \text{for every } H \subseteq \mathcal{E}.$$

Under this condition,

$$H \in \text{Adm}(\mathcal{E}) \iff U_{\mathcal{R}}(H) = \emptyset.$$

Theorem 6.5 is deliberately stronger than a direct canonicity test and deliberately weaker than declaring every structural unsafe activation fatal. It identifies the exact missing object: the behavioral hull of the actual eager target.

6.2 Site-free diagnostic completeness

In finite schedule systems a particularly useful fatality certificate is a continuation whose run no longer depends on any optional eager/lazy choice.

Definition 6.7 (Site-free diagnostic continuation). Let c be a physical configuration. A word

$$z \in \Sigma^*$$

is a *site-free diagnostic continuation* from c if:

- (i) at the configuration reached after every proper prefix of z , including the empty prefix, no optional site in $\mathcal{E}$ is encountered where its eager and lazy actions differ; and
- (ii) after all of z is read, the physical terminal result differs from the semantic terminal result determined by $\Theta(c)$.

Thus the physical run on z is independent of which schedule in $2^{\mathcal{E}}$ is enabled.

Theorem 6.8 (Site-free diagnostic frontier). *Assume $\mathcal{R}$ is a wordwise safe-reset domain for the full family and that every reachable structural unsafe target has a site-free diagnostic continuation:*

$$\forall H \subseteq \mathcal{E}, \forall c \in U_{\mathcal{R}}(H), \exists z_c \text{ site-free diagnostic from } c.$$

Then

$$H \in \text{Adm}(\mathcal{E}) \iff U_{\mathcal{R}}(H) = \emptyset \quad \forall H \subseteq \mathcal{E}.$$

In particular, the reset domain is activation-tight on the whole finite schedule family.

Proof. If $U_{\mathcal{R}}(H) = \emptyset$, the safe-reset certificate gives $H \in \text{Adm}(\mathcal{E})$.

Conversely, suppose

$$c \in U_{\mathcal{R}}(H)$$

and let u be a reaching prefix. By hypothesis, c has a site-free diagnostic z_c . Because that continuation encounters no optional eager/lazy distinction, it has the same physical run under the current schedule H as under any other schedule. Its terminal disagreement therefore makes

$$uz_c$$

a counterexample for H . Hence $H \notin \text{Adm}(\mathcal{E})$. $\square$

Remark 6.9. Site-free diagnostic completeness is sufficient, not assumed universal. A structurally unsafe target may in principle be behaviorally safe, or its fatality may depend on later optional choices. The theorem applies exactly when the schedule-independent diagnostics have been proved.

6.3 Fatal causality

Safe-reset theory identifies fatal targets; causal scheduling asks which optional choices make those targets reachable.

Definition 6.10 (Fatal causal support). Let $b \in \mathcal{E}$ and

$$K \subseteq \mathcal{E} \setminus \{b\}.$$

We write

$$\boxed{K \rightsquigarrow_{\text{fatal}} b}$$

if the background choices in K make reachable a b -source at which taking the eager b -action produces a physical target with a fatal continuation for the schedule $K \cup \{b\}$. When the continuation is site-free, its post-source behavior is schedule independent. Robustness under larger schedules additionally requires that the reaching prefix itself be unchanged by the added optional choices.

Proposition 6.11 (Fatal causality implies conflict). *If*

$$K \rightsquigarrow_{\text{fatal}} b,$$

then

$$\boxed{K \cup \{b\} \notin \text{Adm}(\mathcal{E}).}$$

Proof. Concatenate a prefix reaching the fatal b -source with its finite distinguishing continuation. $\square$

The direction of the arrow records generation of a dangerous exposure, not merely co-occurrence in a bad schedule. In particular, essential participation of two sites in a counterexample does not by itself determine which one causally enabled the failure of the other.

7 Schedule conflict structure

The fatal frontier is a dynamical object. For a finite optional-site family, robust fatality also has a combinatorial shadow.

7.1 Main Theorem VI: from fatal supports to independence complexes

Definition 7.1 (Minimal bad support). A schedule

$$J \subseteq \mathcal{E}$$

is a *minimal bad support* if

$$J \notin \text{Adm}(\mathcal{E})$$

but every proper subset of J is admissible. Let

$$\mathcal{J}_{\min}$$

denote the family of all minimal bad supports.

Lemma 7.2 (Minimal-conflict participation). *If J is a minimal bad support, then every counterexample word for J uses every optional site in J at least once.*

Proof. If a counterexample run never uses some site $e \in J$, then disabling e does not change that run. The same word would therefore be a counterexample for $J \setminus \{e\}$, contradicting minimality. $\square$

Definition 7.3 (Extension-stable fatal witness). A counterexample word w for a bad schedule J is *extension-stable* if it remains a counterexample for every

$$K \supseteq J.$$

A sufficient condition is that along the J -run of w , every site in $\mathcal{E} \setminus J$ is either not encountered or has identical eager and lazy actions at every encountered configuration.

Theorem 7.4 (Robust fatal supports give an independence complex). *Assume every minimal bad support*

$$J \in \mathcal{J}_{\min}$$

has an extension-stable fatal witness. Then badness is upward closed, admissibility is downward closed, and

$$\boxed{\text{Adm}(\mathcal{E}) = \{H \subseteq \mathcal{E} : \nexists J \in \mathcal{J}_{\min} \text{ with } J \subseteq H\} = \text{Ind}(\mathcal{J}_{\min}).}$$

Here $\text{Ind}(\mathcal{J}_{\min})$ denotes the simplicial complex of subsets avoiding every forbidden support in $\mathcal{J}_{\min}$.

Proof. Let J be a minimal bad support and let w_J be an extension-stable witness. By definition, w_J fails for every schedule containing J . Thus every superset of every minimal bad support is bad.

Conversely, if H is bad, finiteness of $\mathcal{E}$ lets us delete sites until an inclusion-minimal bad subset $J \subseteq H$ is reached. Hence H contains a member of $\mathcal{J}_{\min}$. Therefore the bad schedules are exactly the supersets of the minimal bad supports, and the displayed formula follows. $\square$

Corollary 7.5 (Finite diagnostic basis). *For finite $\mathcal{E}$, there is a finite exact regression basis consisting of one distinguishing word for each bad schedule. Under the hypotheses of Theorem 7.4, one extension-stable witness per minimal bad support suffices.*

7.2 A structural two-victim fork

The following template isolates the schedule geometry that later appears in the centered 3:2, 5:2, and odd- p :2 families. Its statement contains no counting residual or kernel arithmetic.

Theorem 7.6 (Reset-ray fork template). *Let*

$$\mathcal{E} = \{A, B, C, D, E, F\}$$

be six optional sites in an exact deterministic lift. Assume:

(i) *the D -off subfamily*

$$\mathfrak{H}_0 = 2^{\{A, B, C, E, F\}}$$

has a wordwise safe-reset domain $\mathcal{R}^{(0)}$ and every schedule in $\mathfrak{H}_0$ preserves it;

(ii) *the subfamily with D enabled and A, B disabled,*

$$\mathfrak{H}_D = \{H \subseteq \mathcal{E} : D \in H, A, B \notin H\},$$

has a wordwise safe-reset domain $\mathcal{R}^{(D)}$ and every schedule in $\mathfrak{H}_D$ preserves it. The two reset domains need not be equal;

(iii) *enabling D makes reachable a new observation-fiber source for A and a new observation-fiber source for B that are absent when D is disabled; and*

(iv) *for each victim $V \in \{A, B\}$ there is a complete extension-stable fatal witness w_V whose reaching prefix encounters no optional action other than D before the eager V -step, and whose continuation after that step is site-free. Equivalently, the D -induced dangerous source remains reachable under arbitrary additions of the other optional sites, and the resulting eager V -target has a schedule-independent fatal continuation.*

Then

$$\boxed{\text{Adm}(\mathcal{E}) = \text{Ind}\{\{A, D\}, \{B, D\}\}.$$

Consequently exactly 40 of the 64 schedules are admissible, and the minimal directed fatal structure is the fork

$$\boxed{D \longrightarrow A, \quad D \longrightarrow B.}$$

Proof. By (i) and Proposition 6.2, all $2^5 = 32$ schedules with D disabled are admissible. By (ii), the same certificate applies to all schedules with D enabled and both A, B disabled; the free choices of C, E, F give another $2^3 = 8$ schedules. Thus every schedule avoiding both pairs $\{A, D\}$ and $\{B, D\}$ is admissible.

Condition (iii) gives the directed exposure mechanism. By (iv), the complete witness for each victim has a reaching prefix that is unchanged by every additional optional choice and a site-free fatal continuation after the victim step. Hence the two witnesses are extension-stable counterexamples for every schedule containing $\{A, D\}$ or $\{B, D\}$, respectively. Therefore every schedule containing either forbidden pair is bad, and the displayed classification follows. $\square$

7.3 Centered baseline used in the weighted-balance case studies

For completeness we record the real-time centered transition schema used in the two weighted-balance case studies below. This subsection is an instantiated testbed for the abstract theory, not a defining hypothesis of it: the fiber, locality, debt, safety, causality, synthesis, and verification theorems above apply to arbitrary deterministic semantic systems satisfying their stated assumptions. Fix coprime positive p, q and put

$$m = p + q, \quad h = \left\lfloor \frac{m-1}{2} \right\rfloor.$$

For a pair (x, y) define its weight

$$W(x, y) = px - qy.$$

Lemma 7.7 (Primitive-kernel minimum demand). *For every $r \in \mathbb{Z}$ there is a unique nonnegative pair*

$$\mu(r) = (x_r, y_r)$$

of weight r minimizing $x + y$. If (x, y) is any other nonnegative pair of weight r , then

$$(x, y) = \mu(r) + k(q, p)$$

for a unique integer $k \geq 0$.

Proof. If two integer pairs have the same weight, then

$$p(x - x') = q(y - y').$$

Coprimality gives

$$(x - x', y - y') = k(q, p)$$

for some $k \in \mathbb{Z}$. Bézout's identity gives at least one integer solution for each r , and adding a sufficiently large multiple of (q, p) makes both coordinates nonnegative. The admissible values of k therefore form a half-line of integers; its first element gives a unique nonnegative pair minimal in the (q, p) direction. Every further step increases $x + y$ by $p + q = m$, proving the claim. $\square$

The centered canonical word, with stack top on the left, is

$$\eta(x, y) = \begin{cases} (10)^y 0^{x-y}, & x \geq y, \\ (10)^x 1^{y-x}, & y > x. \end{cases}$$

The control states are

$$D_{-h}, \dots, D_0 = C, \dots, D_h.$$

For a work-stack word S , let

$$\mathbf{p}(S) = (|S|_0, |S|_1)$$

and define the centered semantic residual

$$\rho_{\text{cent}}(D_d, S) = W(\mathbf{p}(S)) + dm.$$

Thus a canonical center configuration $(C, \eta(\mu(r)))$ has semantic residual r . In the notation of the abstract theory, set

$$\Theta_{\text{cent}} = \rho_{\text{cent}}, \quad \sigma_{\text{cent}}(r) = (C, \eta(\mu(r))),$$

with the bottom marker understood. Then

$$\boxed{\Theta_{\text{cent}}\sigma_{\text{cent}} = \text{id}_{\mathbb{Z}}.}$$

During ordinary binary processing the baseline rules are as follows. We write ε for an empty work stack above the bottom marker Z ; transition rules display Z explicitly when a bottom WRITE is taken. A canonical pair (x, y) is identified with the center configuration $(\mathbf{C}, \eta(x, y))$ (with Z suppressed). The centered recognizer accepts exactly when it terminates in $\mathbf{C}$ with no work-stack symbol above Z .

MATCH. For $z \in \{0, 1\}$,

$$(D_d, zT) \xrightarrow{z} (D_d, T).$$

Interior SEARCH.

$$d < h : \quad (D_d, 0T) \xrightarrow{1} (D_{d+1}, T),$$

$$d > -h : \quad (D_d, 1T) \xrightarrow{0} (D_{d-1}, T).$$

Boundary WRITE. Put

$$U_+ = \eta(\mu((h+1)m)), \quad U_- = \eta(\mu(-(h+1)m)).$$

Then

$$(D_h, 0T) \xrightarrow{1} (\mathbf{C}, U_+T), \quad (D_{-h}, 1T) \xrightarrow{0} (\mathbf{C}, U_-T).$$

Bottom WRITE. With bottom marker Z ,

$$(D_d, Z) \xrightarrow{1} (\mathbf{C}, \eta(\mu(dm+q))Z),$$

$$(D_d, Z) \xrightarrow{0} (\mathbf{C}, \eta(\mu(dm-p))Z).$$

Proposition 7.8 (Centered residual exactness). *Every baseline transition satisfies*

$$\boxed{\rho_{\text{cent}}(\delta_0 c) = \rho_{\text{cent}}(c) - p, \quad \rho_{\text{cent}}(\delta_1 c) = \rho_{\text{cent}}(c) + q.}$$

Hence the centered physical dynamics is an exact lift of the integer residual system on every configuration generated by these rules.

Proof. A MATCH on input 0 removes one stack zero and changes W by $-p$; a MATCH on input 1 removes one stack one and changes W by $+q$. A positive SEARCH on input 1 removes a zero, changing W by $-p$, and increases the control contribution by $m = p + q$, for net change q . A negative SEARCH on input 0 removes a one, changing W by $+q$, and decreases the control contribution by m , for net change $-p$.

At the positive boundary, the source contribution of the visible zero and control is $p + hm$, while the target WRITE block has weight $(h+1)m$; the change is q . At the negative boundary, the source contribution of the visible one and control is $-q - hm$, while the target block has weight $-(h+1)m$; the change is $-p$. Finally, the two bottom WRITES replace residual dm by $dm + q$ or $dm - p$ by definition of μ . These exhaust the baseline rules. $\square$

Proposition 7.9 (Centered canonical-episode terminal correctness). *Suppose an episode starts at a canonical center configuration*

$$(\mathbf{C}, \eta(\mu(r_0)))$$

and, after that reset, reaches an endpoint (D_d, S) using only MATCH and interior SEARCH transitions. Then

$$\boxed{\rho_{\text{cent}}(D_d, S) = 0 \iff (D_d, S) = (\mathbf{C}, \varepsilon).}$$

Consequently, any schedule subfamily for which every reset event returns to a canonical center representative has a wordwise safe-reset certificate: if all reset targets remain inside that canonical reset repertoire, terminal acceptance agrees with semantic residual zero.

Proof. The reverse implication follows immediately from the residual definition. For the forward direction, write

$$\mu(r_0) = (x, y).$$

Between the reset and the endpoint, every MATCH or interior SEARCH consumes exactly one pre-existing work-stack cell. If the intervening input segment contains u zeros and v ones, then

$$u + v \leq x + y.$$

If the endpoint residual is zero, residual exactness gives

$$pu - qv = r_0.$$

Thus (u, v) is a nonnegative completion of residual r_0 . By minimality of $\mu(r_0)$, every such completion has length at least $x + y$. Hence

$$u + v = x + y.$$

The entire canonical work stack has therefore been consumed, so S is empty. The residual equation then reduces to

$$dm = 0,$$

whence $d = 0$ and the control is $\mathbf{C}$.

For the consequence, decompose any word-run at its reset events. Immediately after each reset, the configuration is by hypothesis the canonical center representative of its exact semantic residual. Between consecutive resets the proposition applies. Induction over the episodes therefore shows that a prefix has semantic residual zero exactly when its physical endpoint is the accepting centered configuration. $\square$

There are no ε -repairs during binary-input processing. The optional eager sites below replace selected baseline actions while preserving the same semantic successor.

7.4 Centered 3:2 case study

We record one complete finite instance because it shows how the abstract objects above arise from a concrete real-time LIFO policy.

For

$$p = 3, \quad q = 2,$$

the centered construction has

$$m = p + q = 5, \quad h = 2$$

and control states

$$D_{-2}, D_{-1}, C, D_1, D_2.$$

Its natural six eager sites are indexed by

$$(d, a, X),$$

where d is the centered control index, a the next input symbol, and X the visible top symbol. Their replacement words are:

site	(d, a, X)	γ
A	$(-2, 0, 0)$	11111
B	$(-2, 1, 0)$	10111
C	$(-2, 1, 1)$	11111
D	$(1, 0, 0)$	10100
E	$(1, 1, 0)$	10000
F	$(2, 0, 0)$	10000

Lemma 7.10 (Exact 3:2 finite reset certificate). *Let*

$$\mathcal{R}_0^{(3,2)} = \{(2, 2)\} \cup \{(0, y) : y \geq 4\} \cup \{(1, y) : y \geq 3\} \cup \{(x, 1) : x \geq 3\} \cup \{(x, 0) : x \geq 4\},$$

and

$$\mathcal{R}_D^{(3,2)} = \mathcal{R}_0^{(3,2)} \cup \{(x, 2) : x \geq 3\}.$$

Before the first reset event of an episode started from one of these canonical pairs, the relevant eager-source shapes are

local context	$\mathcal{R}_0^{(3,2)}$	$\mathcal{R}_D^{(3,2)}$	
$D_{-2}, \text{ top } 0$	0	0^k	$(k \geq 1).$
$D_{-2}, \text{ top } 1$	1^k	1^k	
$D_1, \text{ top } 0$	0^k	0^k	
$D_2, \text{ top } 0$	0^k	0^k	

The bottom-WRITE targets are

d	0	1
-2	(1, 8)	(0, 4)
-1	(0, 4)	(1, 3)
0	(1, 3)	(2, 2)
1	(2, 2)	(3, 1)
2	(3, 1)	(4, 0),

and every one lies in $\mathcal{R}_0^{(3,2)}$. For the remaining reset events, writing the unary source tail as $t \geq 0$, the target pairs are

reset action	canonical target pair
positive boundary WRITE	$(t + 5, 0)$
negative boundary WRITE	$(1, t + 9)$
A	$(0, 5)$
B	$(1, 4)$
C	$(0, t + 5)$
D	$(t + 3, 2)$
E	$(t + 4, 1)$
F	$(t + 4, 1).$

Consequently:

- (a) with D disabled, every reset event returns to $\mathcal{R}_0^{(3,2)}$, independently of the eager/lazy choices at A, B, C, E, F ;
- (b) with D enabled and A, B disabled, every reset event returns to $\mathcal{R}_D^{(3,2)}$, independently of the choices at C, E, F .

Moreover, the new D -ray is exactly what enlarges the negative-boundary top-0 source from the singleton 0 to the unary family 0^k .

Proof. The canonical words in $\mathcal{R}_0^{(3,2)}$ are

$$1010, \quad 1^y, \quad 101^{y-1}, \quad 10^x, \quad 0^x,$$

in the five displayed pair families, while

$$\eta(x, 2) = 1010^{x-1}$$

for the added D -ray. Before the first reset, MATCH and interior SEARCH only delete the visible top cell, so every current stack is a suffix of its starting canonical word. Direct suffix inspection gives the four source rows; in particular, only 1010 can reach D_{-2} with a zero still visible without using the D -ray, and it leaves exactly the final zero.

For bottom WRITES the table is the direct evaluation of $\mu(5d - 3)$ and $\mu(5d + 2)$. For all other resets, the centered boundary blocks and the six replacement words give, respectively,

$$0^{t+5}, \quad 101^{t+8}, \quad 1^5, \quad 101^3, \quad 1^{t+5}, \quad 1010^{t+2}, \quad 10^{t+4}.$$

These are precisely the canonical encodings of the target pairs in the second table. Every row except D lies in $\mathcal{R}_0^{(3,2)}$, while the D row is exactly the additional ray of $\mathcal{R}_D^{(3,2)}$. This proves both closure statements and the claimed source-fiber expansion. $\square$

Theorem 7.11 (Exact 3:2 schedule classification). *For the six-site centered 3:2 family,*

$$\text{Adm}_{3,2} = \{H : \{A, D\} \not\subseteq H, \{B, D\} \not\subseteq H\}.$$

Hence

$$40 \text{ schedules are admissible and } 24 \text{ are inadmissible.}$$

The minimal fatal graph is

$$D \rightarrow A, \quad D \rightarrow B.$$

The two facets are

$$\{A, B, C, E, F\}, \quad \{C, D, E, F\},$$

so the admissible schedule complex is non-pure and the schedule poset is not a lattice. Its independence polynomial is

$$(1 + x)^3(1 + 3x + x^2).$$

Proof. Lemma 7.10(a) gives a closed canonical reset induction for every schedule with D disabled. By Proposition 7.9, $\mathcal{R}_0^{(3,2)}$ is therefore a wordwise safe-reset domain for this schedule subfamily. Hence all $2^5 = 32$ schedules with $D = 0$ are admissible.

Lemma 7.10(b) gives the second safe regime. When D is enabled and $A = B = 0$, the only new canonical reset family is

$$\{(x, 2) : x \geq 3\},$$

and C, D, E, F preserve the enlarged repertoire $\mathcal{R}_D^{(3,2)}$. Proposition 7.9 therefore makes $\mathcal{R}_D^{(3,2)}$ a wordwise safe-reset domain for this second subfamily. Hence the $2^3 = 8$ schedules with

$$D = 1, \quad A = B = 0$$

are admissible. This proves sufficiency for exactly the 40 schedules avoiding both pairs $\{A, D\}$ and $\{B, D\}$.

For necessity, the new ray makes longer D_{-2} -top-0 sources reachable, as identified in Lemma 7.10. Two complete balanced witnesses are

$$\boxed{w_B = 1111000011}, \quad \boxed{w_A = 011111110000011}.$$

Their critical portions are

branch	reaching prefix	victim step	diagnostic
$D \rightarrow B$	11110000 : $(D_{-2}, 00)$	1 : $(C, 101110)$	1 : $(C, 01110)$
$D \rightarrow A$	011111110000 : $(D_{-2}, 000)$	0 : $(C, 1111100)$	11 : $(C, 11100)$.

In the first row the earlier eager D -step occurs on the prefix 11110; in the second it occurs on the prefix 01111110. Before the victim step the only eager optional site used is therefore D , and after that step the listed diagnostic uses only center-state baseline transitions. Consequently the two complete words are extension-stable under arbitrary additions of C, E, F and of the other victim site.

Both terminal configurations in the last column have semantic residual zero but a nonempty center stack, so they are nonaccepting under the fixed centered acceptance convention. Hence every schedule containing $\{B, D\}$ or $\{A, D\}$ is bad. The hypotheses of Theorem 7.6 are now verified explicitly, giving the classification. The facets and independence polynomial follow from the two forbidden edges. $\square$

7.5 An infinite exact family

The same structural fork persists for an infinite parameter family. Here we give the parameter proof in full, rather than treating the finite 3:2 and 5:2 instances as isolated calculations.

Fix

$$p = 2h - 1, \quad q = 2, \quad h \geq 2,$$

so that

$$m = p + q = 2h + 1.$$

For $t \geq 1$ put

$$\epsilon_t = t \bmod 2, \quad k_t = \left\lceil \frac{t}{2} \right\rceil, \quad \beta_t = t + p k_t,$$

and define

$$\Gamma_t = \eta(\epsilon_t, \beta_t).$$

The natural six eager sites are

$$\begin{aligned}
A &: (D_{-h}, 0T) \xrightarrow{0} (\mathbb{C}, \Gamma_h T), \\
B &: (D_{-h}, 0T) \xrightarrow{1} (\mathbb{C}, \Gamma_{h-1} T), \\
C &: (D_{-h}, 1T) \xrightarrow{1} (\mathbb{C}, \Gamma_h T), \\
D &: (D_{h-1}, 0T) \xrightarrow{0} (\mathbb{C}, (10)^h 0T), \\
E &: (D_{h-1}, 0T) \xrightarrow{1} (\mathbb{C}, (10)^{h-1} 000T), \\
F &: (D_h, 0T) \xrightarrow{0} (\mathbb{C}, (10)^{h-1} 000T).
\end{aligned}$$

Lemma 7.12 (Odd- p :2 arithmetic catalogue). *For $p = 2h - 1$, $q = 2$, and $m = 2h + 1$:*

(i) *for every $t \geq 1$,*

$$\boxed{\mu(-tm) = (\epsilon_t, \beta_t);}$$

(ii) *for $0 \leq j \leq h$,*

$$\boxed{\mu(jm + 2) = (j + 2, p - j - 1);}$$

(iii) *for $1 \leq t \leq h + 1$,*

$$\boxed{\mu(-tm + 2) = (\epsilon_t, \beta_t - 1);}$$

(iv) *the positive outer boundary demand is*

$$\boxed{\mu((h + 1)m) = (h + 3, h - 2).}$$

Consequently the complete family of canonical baseline reset pairs can be written

$$\begin{aligned}
\mathcal{R}_h^{\text{base}} &= \{\mu(jm + 2) : -h - 1 \leq j \leq h\} \\
&\cup \{(x, h - 2) : x \geq h + 3\} \\
&\cup \{(\epsilon_{h+1}, y) : y \geq \beta_{h+1}\}.
\end{aligned}$$

Proof. For $t = 2r$,

$$(0, \beta_t) = (0, 2r + pr)$$

has weight

$$-2(2r + pr) = -2r(p + 2) = -tm.$$

For $t = 2r + 1$,

$$(1, \beta_t) = (1, 2r + 1 + p(r + 1))$$

has weight

$$p - 2(2r + 1 + p(r + 1)) = -(2r + 1)(p + 2) = -tm.$$

In both cases the first coordinate is below the kernel step $q = 2$, so the pair is kernel-reduced and therefore equals $\mu(-tm)$. This proves (i).

For (ii), the displayed pair has weight

$$p(j + 2) - 2(p - j - 1) = j(p + 2) + 2 = jm + 2,$$

and its second coordinate is strictly below p , so it is kernel-reduced. Part (iii) follows from (i) because decreasing the second coordinate by one adds 2 to the weight; the first coordinate remains below 2. Finally,

$$p(h+3) - 2(h-2) = (h+1)(2h+1) = (h+1)m,$$

and $h-2 < p$, proving (iv).

A bottom input 1 from state D_d has demand $dm+2$, while a bottom input 0 has demand

$$dm - p = (d-1)m + 2.$$

As $-h \leq d \leq h$, the union of bottom-WRITE demands is exactly

$$jm + 2, \quad -h-1 \leq j \leq h.$$

At the positive outer boundary, appending a unary-zero tail to $\eta(h+3, h-2)$ yields the ray $(x, h-2)$, $x \geq h+3$. At the negative outer boundary, appending a unary-one tail to Γ_{h+1} yields (ϵ_{h+1}, y) with $y \geq \beta_{h+1}$. These are all baseline reset families. $\square$

Define the two candidate reset repertoires

$$\begin{aligned} \mathcal{R}_{-D,h} = & \mathcal{R}_h^{\text{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \\ & \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h-1) : x \geq h+2\}, \end{aligned}$$

and

$$\begin{aligned} \mathcal{R}_{D,h} = & \mathcal{R}_h^{\text{base}} \cup \{(\epsilon_h, y) : y \geq \beta_h\} \\ & \cup \{(x, h-1) : x \geq h+2\} \cup \{(x, h) : x \geq h+1\}. \end{aligned}$$

Every pair in either repertoire has minority size at most h .

Lemma 7.13 (Odd- p :2 source geometry). *Consider an episode that starts at a canonical center pair in one of the repertoires above and stops at its first reset event.*

(i) *If the starting pair lies in $\mathcal{R}_{-D,h}$ and the machine reaches D_{-h} with visible top 0, then the complete remaining stack is exactly*

$$\boxed{0.}$$

(ii) *If the starting pair lies in $\mathcal{R}_{D,h}$ and the machine reaches D_{-h} with visible top 0, the remaining stack is unary:*

$$\boxed{0^n, \quad n \geq 1.}$$

The lengths $n \geq 2$ can arise only from the additional D -ray $\{(x, h) : x \geq h+1\}$.

(iii) *In either repertoire,*

$$D_{-h}, \text{ top } 1 \implies \text{stack } 1^n,$$

$$D_{h-1}, \text{ top } 0 \implies \text{stack } 0^n,$$

and

$$D_h, \text{ top } 0 \implies \text{stack } 0^n$$

for some $n \geq 1$.

Proof. Before the first reset event, MATCH and interior SEARCH only delete the visible top cell, so every remaining stack is a suffix of the canonical reset word from which the episode started.

For (i), reaching control D_{-h} requires at least h more negative SEARCHes than positive SEARCHes; hence at least h one-cells have already been popped. If a zero is still visible, the original canonical pair must therefore contain at least h zeros and at least h ones. Inspect the families in $\mathcal{R}_{-D,h}$. Negative bottom and negative outer families, as well as the $A/B/C$ families, have first coordinate at most 1; the E/F family has second coordinate $h - 1$; the positive outer family has second coordinate $h - 2$. A positive bottom pair has the form

$$(j + 2, 2h - 2 - j).$$

The simultaneous inequalities

$$j + 2 \geq h, \quad 2h - 2 - j \geq h$$

force $j = h - 2$, giving the unique pair (h, h) . Its canonical word is $(10)^h$. Since it contains only h one-cells, net control $-h$ forces all of them to be negative SEARCHes and forbids any positive SEARCH. The intervening zeros are MATCHed, leaving exactly the final zero. This proves (i).

The only additional family in $\mathcal{R}_{D,h}$ capable of leaving a longer zero suffix at D_{-h} is

$$(x, h), \quad x \geq h + 1,$$

whose canonical word is $(10)^h 0^{x-h}$. After the h required negative SEARCHes, its remaining suffix is unary zero. This proves (ii).

For the first and third guards in (iii), suppose instead that the visible boundary symbol had the opposite symbol later in the same suffix. At D_{-h} with top 1, at least h earlier one-cells have contributed to the net negative search count, so a later zero would force at least $h + 1$ zeros and $h + 1$ ones in the starting pair. Similarly, at D_h with top 0, a later one would force minority at least $h + 1$. Both contradict the minority- h bound of the two repertoires.

For D_{h-1} with top 0, a later one would require at least h zeros and $h + 1$ ones. None of the listed families has that shape: negative and $A/B/C$ families have at most one zero; positive outer pairs have only $h - 2$ ones; E/F pairs have $h - 1$ ones; the D -ray has exactly h ones; and a positive bottom pair would require simultaneously

$$j + 2 \geq h, \quad 2h - 2 - j \geq h + 1,$$

that is $j \geq h - 2$ and $j \leq h - 3$, impossible. Hence the remaining suffix is unary zero. $\square$

Lemma 7.14 (Odd- $p:2$ reset closure). *The following two schedule regimes are reset-closed.*

- (a) *If D is disabled, every baseline reset and every enabled eager action among A, B, C, E, F sends an episode started in $\mathcal{R}_{-D,h}$ to a canonical pair again in $\mathcal{R}_{-D,h}$.*
- (b) *If D is enabled and A, B are disabled, every baseline reset and every enabled eager action among C, D, E, F sends an episode started in $\mathcal{R}_{D,h}$ to a canonical pair again in $\mathcal{R}_{D,h}$.*

Proof. All baseline bottom and boundary WRITES land in $\mathcal{R}_h^{\text{base}}$ by Lemma 7.12. It remains to inspect eager resets. By Lemma 7.13, write their unary source stacks as shown below, with $n \geq 1$:

site	source stack	canonical target pair
A	0	(ϵ_h, β_h)
B	0	$(\epsilon_{h-1}, \beta_{h-1})$
C	1^n	$(\epsilon_h, \beta_h + n - 1)$
D	0^n	$(n + h, h)$
E	0^n	$(n + h + 1, h - 1)$
F	0^n	$(n + h + 1, h - 1).$

For C , appending the tail 1^{n-1} to Γ_h simply extends its final unary-one run. For D ,

$$(10)^h 0 0^{n-1} = \eta(n+h, h),$$

while for E and F ,

$$(10)^{h-1} 000 0^{n-1} = \eta(n+h+1, h-1).$$

Thus every row is exactly canonical.

In regime (a), Lemma 7.13(i) makes the A/B source the one-cell stack 0. The A target belongs to the C -ray $\{(\epsilon_h, y) : y \geq \beta_h\}$, the B target is the listed singleton, and the C, E, F targets lie in their corresponding rays. Hence $\mathcal{R}_{-D, h}$ is closed.

In regime (b), the D row generates exactly

$$\{(x, h) : x \geq h+1\},$$

while C, E, F return to the other listed families. Hence $\mathcal{R}_{D, h}$ is closed. $\square$

Lemma 7.15 (Uniform D -induced fiber expansion). *For every $x \geq h+1$,*

$$\boxed{(C, \eta(x, h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}).}$$

Consequently D expands the common A/B source fiber from the singleton $\{0\}$ to

$$\boxed{\{0, 00, 000, \dots\}}.$$

Proof. Since

$$\eta(x, h) = (10)^h 0^{x-h},$$

the first $2h-2$ zeros remove $h-1$ complete 10 pairs: every 1 causes a negative SEARCH and every following 0 MATCHes. The final input zero SEARCHes through the last 1, placing the control at D_{-h} and leaving the last alternating zero together with the zero tail:

$$0^{x-h+1}.$$

The D -ray contains every $x \geq h+1$, hence produces every unary-zero length at least two. Lemma 7.13(i) shows that without D only length one is possible. $\square$

Lemma 7.16 (Uniform reachability of the length-three frontier). *For every $h \geq 2$, there is a word u_h that reaches*

$$\boxed{(D_{-h}, 000)}$$

from the initial centered configuration and encounters no eager optional site other than D .

Let

$$v_h = \begin{cases} 1^{(h-1)m/2}, & h \text{ odd}, \\ 0 1^{2h+(h-2)m/2}, & h \text{ even}. \end{cases}$$

Then

$$\boxed{u_h = v_h 1^{2h-1} 0^{2h}}.$$

Proof. We first establish a two-state increment block. If

$$0 \leq d \leq h - 2$$

and the machine is at (D_d, ε) , then the first symbol of 1^m bottom-WRITEs the pair

$$\mu(dm + 2) = (d + 2, p - d - 1).$$

Its canonical word has length

$$(d + 2) + (p - d - 1) = p + 1 = 2h.$$

The remaining $2h$ input ones remove this entire word. Each of its $d + 2$ zeros is read as a positive SEARCH mismatch, so

$$\boxed{(D_d, \varepsilon) \xrightarrow{1^m} (D_{d+2}, \varepsilon)}.$$

If h is odd, $(h - 1)/2$ such blocks take D_0 to D_{h-1} , proving the first formula for v_h . If h is even, the prefix 01^{2h} first bottom-WRITEs

$$\mu(-p) = (1, p), \quad \eta(1, p) = 101^{p-1}.$$

The following $2h = p + 1$ ones remove this word and perform exactly one positive SEARCH, ending at (D_1, ε) . A further $(h - 2)/2$ blocks 1^m reach (D_{h-1}, ε) , proving the even formula.

From (D_{h-1}, ε) , the first symbol of 1^{2h-1} bottom-WRITEs

$$\mu((h - 1)m + 2) = (h + 1, h - 1),$$

whose canonical word is

$$(10)^{h-1}00.$$

The remaining $2h - 2$ ones remove exactly the alternating prefix and leave

$$(D_{h-1}, 00).$$

The next input 0 invokes D , yielding

$$(\mathbf{C}, (10)^h 00).$$

Finally 0^{2h-1} removes $h - 1$ complete 10 pairs and SEARCHes through the last 1, leaving

$$(D_{-h}, 000).$$

Along the increment blocks the control never presents an eager source; in the 1^{2h-1} segment the last positive SEARCH reaches D_{h-1} only after its visible zero has been removed. The next symbol is precisely the eager D -step, and the final zero segment reaches D_{-h} only on its last symbol. Hence no site among A, B, C, E, F is encountered before the endpoint. $\square$

Lemma 7.17 (Uniform kernel-suffix fatality). *For every $t \geq 1$, Γ_t has the suffix 1^p . Therefore there is a unique word z_t such that*

$$\boxed{\Gamma_t = z_t 1^p}.$$

At the common source $(D_{-h}, 000)$, eager A followed by input z_h and eager B followed by input z_{h-1} both reach

$$\boxed{(\mathbf{C}, 1^p 00)}.$$

The two continuations are site-free fatal diagnostics.

Proof. If $t = 2r$ is even, then

$$\Gamma_t = 1^{\beta_t}, \quad \beta_t = r(p+2) = rm \geq m > p.$$

If $t = 2r + 1$ is odd, then

$$\Gamma_t = 10 1^{\beta_t-1}, \quad \beta_t - 1 = p + rm \geq p.$$

Thus Γ_t ends in at least p consecutive ones.

At source $(D_{-h}, 000)$ the victim rule removes the visible first zero and leaves tail 00. Hence

$$A : (D_{-h}, 000) \xrightarrow{0} (\mathbb{C}, \Gamma_h 00) = (\mathbb{C}, z_h 1^p 00),$$

and

$$B : (D_{-h}, 000) \xrightarrow{1} (\mathbb{C}, \Gamma_{h-1} 00) = (\mathbb{C}, z_{h-1} 1^p 00).$$

Reading z_h or z_{h-1} from the center consists solely of MATCHes and therefore reaches $(\mathbb{C}, 1^p 00)$ without visiting any optional site.

The remaining stack has two zeros and p ones, so its weight is

$$p \cdot 2 - 2 \cdot p = 0.$$

It is nevertheless nonempty, and the centered acceptance convention accepts only the empty center stack at semantic residual zero. Thus both continuations are fatal. $\square$

Theorem 7.18 (Odd- $p:2$ fatal-fork family). *For every odd $p \geq 3$ in the centered $p:2$ six-site family,*

$$\text{Adm}_{p,2} = \text{Ind}\{\{A, D\}, \{B, D\}\}.$$

Equivalently,

$$H \in \text{Adm}_{p,2} \iff \neg(A \wedge D) \wedge \neg(B \wedge D).$$

Thus every member of the family has exactly 40 admissible and 24 inadmissible schedules, with facets

$$\{A, B, C, E, F\}, \quad \{C, D, E, F\},$$

and fatal fork

$$D \rightarrow A, \quad D \rightarrow B.$$

Proof. Lemma 7.14(a) proves that every schedule with D disabled has a closed canonical reset induction. By Proposition 7.9, the repertoire $\mathcal{R}_{-D,h}$ is therefore a wordwise safe-reset domain for that schedule subfamily, giving all $2^5 = 32$ schedules with $D = 0$. Lemma 7.14(b) and the same proposition make $\mathcal{R}_{D,h}$ a wordwise safe-reset domain when D is enabled and $A = B = 0$, contributing another $2^3 = 8$ schedules.

Lemma 7.15 identifies the directed exposure mechanism. By Lemma 7.16, the same reaching prefix u_h reaches $(D_{-h}, 000)$ and encounters only eager D . At that source, Lemma 7.17 gives the complete counterexample words

$$w_A^{(h)} = u_h 0 z_h, \quad w_B^{(h)} = u_h 1 z_{h-1}.$$

Their suffixes after the victim step are site-free, and their common reaching prefix encounters no optional site except D . Thus both witnesses are extension-stable under arbitrary additions of C, E, F and of the other victim site. Hence $\{A, D\}$ and $\{B, D\}$ are robust minimal bad supports.

The hypotheses of Theorem 7.6 are now verified for every $h \geq 2$, yielding the stated classification. $\square$

Remark 7.19 (Why this family matters). The odd- $p:2$ theorem is not used to define the General Theory. Its role is the opposite: it is a parameterized stress test showing that the abstract safe-reset, exposure, and robust-fatality mechanism persists across an infinite family of distinct centered pushdown systems.

8 Section synthesis and forced debt

The previous parts diagnose a fixed representation: they identify when a chosen section is locally realizable, when debt is behaviorally harmless, and when optional normalization choices interact causally. We now reverse the question.

Given the semantic dynamics and an exact physical realization, can a transition-closed canonical section be constructed rather than guessed?

For rooted systems the answer has a simple form. Once one physical representative of the semantic root is fixed, transition closure determines every other section value. The only possible obstruction is path dependence: two semantic paths reaching the same semantic state may arrive at different physical representatives.

8.1 Main Theorem VII-A: rooted path-outcome coherence synthesis

Assume in this subsection that the semantic component under discussion is rooted at m_0 , meaning that every $m \in M$ is reachable from m_0 . Let

$$c_0 \in \Theta^{-1}(m_0)$$

be a candidate physical root representative. We seek a new section

$$\tau : M \rightarrow C$$

of the same exact lift; τ need not coincide with the previously chosen section σ .

Definition 8.1 (Rooted path-outcome fiber). For $m \in M$, define

$$\boxed{\mathcal{H}_{c_0}(m) = \{\delta_w(c_0) : U_w(m_0) = m\}.}$$

Exactness implies

$$\mathcal{H}_{c_0}(m) \subseteq \Theta^{-1}(m).$$

Theorem 8.2 (Rooted path-outcome coherence criterion). *Fix $c_0 \in \Theta^{-1}(m_0)$. The following are equivalent:*

(i) *there exists a transition-closed section τ satisfying*

$$\tau(m_0) = c_0, \quad \delta_a \tau = \tau U_a \quad \forall a \in \Sigma;$$

(ii) *for every semantic state m ,*

$$\boxed{|\mathcal{H}_{c_0}(m)| = 1.}$$

When these conditions hold, the section is unique.

Proof. Assume first that such a section τ exists. If

$$U_w(m_0) = m,$$

then transition closure gives

$$\delta_w(c_0) = \delta_w(\tau(m_0)) = \tau(U_w(m_0)) = \tau(m).$$

Hence every member of $\mathcal{H}_{c_0}(m)$ is the same physical configuration.

Conversely, suppose every path-outcome fiber is a singleton. Define $\tau(m)$ to be its unique element. Exactness gives

$$\Theta\tau(m) = m,$$

so τ is a section. If w reaches m , then

$$\delta_a\tau(m) = \delta_{wa}(c_0).$$

The word wa reaches $U_a(m)$ semantically, so path-outcome coherence gives

$$\delta_{wa}(c_0) = \tau(U_a(m)).$$

Thus $\delta_a\tau = \tau U_a$.

Finally, any transition-closed section extending c_0 must satisfy

$$\tau(m) = \delta_w(c_0)$$

for every word w reaching m . Hence it is unique. $\square$

Corollary 8.3 (One root representative determines the section). *For a rooted semantic system, the values of a transition-closed section are not independent design choices. Once $\tau(m_0) = c_0$ is fixed,*

$$\boxed{\tau(m) = \delta_w(c_0)}$$

for any semantic path w from m_0 to m .

This yields a useful interpretation:

$$\boxed{\text{rooted section synthesis} = \text{elimination of physical path dependence inside semantic fibers.}}$$

8.2 Finite rooted synthesis

Suppose now that M is finite and that a finite set

$$K_0 \subseteq \Theta^{-1}(m_0)$$

of admissible root representatives is given. If the policy restricts which representatives may be canonical, let

$$\text{Can}(m, c)$$

be an effectively decidable admissibility predicate.

Finite propagation procedure. For each $c_0 \in K_0$:

1. assign $S[m_0] = c_0$ and leave every other semantic state unassigned;
2. traverse the rooted semantic transition graph;
3. whenever $S[m] = c$ is assigned and

$$U_a(m) = m',$$

compute

$$c' = \delta_a(c);$$

4. if $S[m']$ is unassigned, set $S[m'] = c'$; if it is already assigned, require

$$S[m'] = c';$$

5. if the physical policy restricts canonical representatives, also require

$$\text{Can}(m', S[m']);$$

6. reject the root candidate on the first failed equality or admissibility test; otherwise, after every semantic edge has been checked, output

$$\tau(m) = S[m].$$

Theorem 8.4 (Finite rooted synthesis). *The finite propagation procedure above is exact relative to the specified root candidate set and representative-admissibility predicate. For each $c_0 \in K_0$, it outputs the unique transition-closed section extending c_0 if such a section exists, and otherwise rejects c_0 .*

With constant-time physical transition, equality, and admissibility tests, the total number of local checks is

$$O(|K_0| |M| |\Sigma|).$$

Proof. For a fixed root candidate, every semantic edge is checked once. Whenever a semantic state is first reached, transition closure forces its physical representative to be the propagated value. A later inconsistent revisit exhibits two paths to the same semantic state with different physical outcomes, violating Theorem 8.2. Conversely, if all revisits agree, every semantic edge satisfies

$$\delta_a(S[m]) = S[U_a(m)],$$

so S is a transition-closed section. The complexity bound follows from the at most $|M||\Sigma|$ labeled semantic edges per root candidate. $\square$

Theorem 8.5 (Short parallel-path obstruction certificate). *Let $n = |M|$. If a root candidate c_0 fails because one semantic state receives two different propagated physical values, then there exist words u, v such that*

$$U_u(m_0) = U_v(m_0)$$

but

$$\delta_u(c_0) \neq \delta_v(c_0),$$

with

$$|u| \leq n, \quad |v| \leq n - 1.$$

Proof. Use a breadth-first semantic spanning tree. Let v be the tree word of the state receiving the conflicting value. Let x be the tree word of the predecessor state on the non-tree edge causing the conflict, and let a be that edge label. Then $u = xa$ and v reach the same semantic state but the propagation test found

$$\delta_u(c_0) \neq \delta_v(c_0).$$

Breadth-first tree words have length at most $n - 1$, hence $|v| \leq n - 1$ and $|u| \leq n$. $\square$

Corollary 8.6 (Counting synthesized sections). *If K_0^{good} is the set of accepted root candidates, then*

$$\boxed{\#\{\text{transition-closed sections rooted in } K_0\} = |K_0^{\text{good}}|}.$$

Example 8.7 (Automatic repair of the 1^* allocation failure). Recall the exact physical lift from Example 3.12. Fix the live root representative

$$c_0 = AZ.$$

The semantic edge

$$L \xrightarrow{0} D$$

and the physical edge

$$AZ \xrightarrow{0} Z$$

force

$$\boxed{\tau(D) = Z}.$$

The dead loops preserve Z , so the synthesis succeeds. The previously chosen representative ABZ is not merely inelegant: once AZ is fixed at the root, it is inconsistent with transition closure. Thus the synthesis algorithm automatically repairs the pure Stage-II allocation obstruction.

8.3 Main Theorem VII-B: symbolic synthesis on an infinite semantic core

The path-outcome coherence criterion itself does not require M to be finite. Finiteness was used only to traverse the semantic graph explicitly. The integer-translation core

$$M = \mathbb{Z}, \quad U_a(r) = r + \omega(a),$$

provides a simple infinite-domain synthesis in which infinitely many semantic states are represented by finitely many local arithmetic schemas.

Let Σ be finite and

$$\omega : \Sigma \rightarrow \mathbb{Z}.$$

For a word $w = a_1 \cdots a_n$, write

$$\omega(w) = \sum_{i=1}^n \omega(a_i).$$

Theorem 8.8 (Modular quotient/remainder synthesis). *Let*

$$W_{\max} = \max_{a \in \Sigma} |\omega(a)|$$

and choose

$$\boxed{B > W_{\max}},$$

for example $B = \max(2, W_{\max} + 1)$.

There exists a deterministic real-time one-counter lift with unary stack symbol X , bottom marker Z , and finite controls

$$Q_0^+, \dots, Q_{B-1}^+, \quad Q_0^-, \dots, Q_{B-1}^-,$$

together with a transition-closed section

$$\zeta_B : \mathbb{Z} \rightarrow C_B$$

defined by

$$\zeta_B(r) = \begin{cases} (Q_d^+, X^n Z), & r = Bn + d, \quad r \geq 0, \quad 0 \leq d < B, \\ (Q_d^-, X^n Z), & -r = Bn + d, \quad r < 0, \quad 0 \leq d < B. \end{cases}$$

The sign/quotient/remainder representation is unique, so ζ_B is injective. Take the invariant policy domain to be

$$C_B = \zeta_B(\mathbb{Z})$$

and define the semantic projection by

$$\Theta_B(\zeta_B(r)) = r.$$

Then

$$\Theta_B \zeta_B = \text{id}_{\mathbb{Z}}.$$

The transition table can be generated effectively so that

$$\delta_a \zeta_B(r) = \zeta_B(r + \omega(a))$$

for every $r \in \mathbb{Z}$ and $a \in \Sigma$. Hence

$$\Theta_B \delta_a = U_a \Theta_B$$

on C_B , so (C_B, δ, Θ_B) is an exact lift. Unused syntactic one-counter configurations may be completed arbitrarily outside this invariant policy domain.

Proof. Fix a and put

$$s = \omega(a), \quad |s| < B.$$

Suppose first that

$$r = Bn + d, \quad n \geq 1, \quad 0 \leq d < B.$$

The visible stack top is X . Put $t = d + s$. Since $|s| < B$, the sign cannot cross zero while at least one full B -block remains, and the quotient height changes by at most one:

condition	new remainder	stack action
$t < 0$	$t + B$	pop one X
$0 \leq t < B$	t	keep height
$t \geq B$	$t - B$	push one X .

This produces exactly the encoding of $r + s$.

For a negative value

$$-r = Bn + d, \quad n \geq 1,$$

the same argument applies to the magnitude after replacing t by

$$t = d - s.$$

Again the quotient changes only by $-1, 0, +1$ and the sign remains negative.

When the bottom marker is visible, the whole magnitude is already determined by finite control: it is d in control Q_d^+ and $-d$ in canonical control Q_d^- . Directly compute $r + s$ and encode it by ζ_B . Since $|s| < B$ and the old magnitude is $< B$, the target uses at most one pushed X .

Thus each combination of sign, remainder, top category, and input symbol determines one legal real-time one-top rule, and every such rule maps the canonical source to the canonical semantic successor. $\square$

Theorem 8.9 (Finite symbolic skeleton and global path-outcome coherence). *The construction of Theorem 8.8 is determined by at most*

$$(4B - 1)|\Sigma|$$

canonical local arithmetic cases. Moreover, for every $r \in \mathbb{Z}$ and every word w ,

$$\delta_w(\zeta_B(r)) = \zeta_B(r + \omega(w)).$$

Hence

$$\omega(u) = \omega(v) \implies \delta_u(\zeta_B(r)) = \delta_v(\zeta_B(r))$$

for all r .

Proof. The canonical local observation schemas are

$$(Q_d^+, X), \quad (Q_d^+, Z), \quad (Q_d^-, X), \quad (Q_d^-, Z) \quad (d \geq 1),$$

so there are at most $4B - 1$ of them. With X visible, the rule depends on the finite remainder and input weight but not on the hidden quotient height; with Z visible, the exact bounded magnitude is known. This proves finite skeleton completeness.

The word identity follows by induction from the one-symbol identity of Theorem 8.8. Equal accumulated weights therefore give equal physical endpoints, which is exactly the path-outcome coherence condition of Theorem 8.2. $\square$

Corollary 8.10 (Zero-weight loop coherence). *If*

$$\omega(w) = 0,$$

then

$$\delta_w(\zeta_B(r)) = \zeta_B(r)$$

for every $r \in \mathbb{Z}$.

Remark 8.11 (Scope and complexity). The explicit rule table has size

$$O(B|\Sigma|).$$

This is output-linear in the finite skeleton and pseudo-polynomial in the numerical weights. If weights are binary encoded, no polynomial bound in their bit length is claimed. The result is also not a search procedure for a section inside an arbitrary pre-existing infinite lift. It is a joint *lift-and-section synthesis* theorem inside a one-counter template. No state-minimality claim is made.

Example 8.12 (Three-letter weighted language). For

$$\omega(a) = 2, \quad \omega(b) = 3, \quad \omega(c) = -5,$$

take $B = 6$. The resulting raw quotient/remainder coordinates are transition-closed even though every block-literal shortest-completion section fails one-top closure as shown in Example 3.13. Thus synthesis does more than verify representation relativity: it constructs an explicit coordinate system in which the former locality debt disappears.

8.4 Partial sections and forced-debt frontiers

Total transition closure is not always possible or desirable. A semantic system may have a large region that admits exact canonical synthesis and a boundary at which finite LIFO observation forces departure from the section.

Definition 8.13 (Partial canonical section). Let

$$G \subseteq M.$$

A partial section is a map

$$\tau_G : G \rightarrow C$$

satisfying

$$\Theta \tau_G = \text{id}_G.$$

It is internally transition closed if

$$\boxed{m \in G, U_a(m) \in G \implies \delta_a \tau_G(m) = \tau_G(U_a(m))}.$$

Theorem 8.14 (Canonical interior). *Let τ_G be internally transition closed. Any physical run started at a canonical point $\tau_G(m)$ remains canonical for as long as its semantic run remains inside G . If a_j is the first input whose semantic successor leaves G , then the exit target need not be canonical, but exactness still gives*

$$\boxed{\Theta(\delta_{a_j}(\tau_G(m_{j-1}))) = U_{a_j}(m_{j-1})}.$$

Proof. Induct over the internal semantic path. Every edge remaining in G is canonical by hypothesis. At the first exit, apply exactness:

$$\Theta \delta_{a_j} \tau_G(m_{j-1}) = U_{a_j} \Theta \tau_G(m_{j-1}) = U_{a_j}(m_{j-1}).$$

□

The next bound strengthens the collapse-capacity theorem from Section 3: it does not assume that the whole section is transition closed.

Theorem 8.15 (Canonical-hit capacity). *Fix an input symbol a , a semantic target m^* , and a source set*

$$\mathcal{S} \subseteq U_a^{-1}(m^*).$$

Suppose a representative map τ is defined on $\mathcal{S} \cup \{m^\}$ and satisfies*

$$\Theta \tau(m) = m \quad (m \in \mathcal{S} \cup \{m^*\}).$$

Suppose also that the physical policy has finite local observation set Ω_a and tail-injective hidden updates.

Define

$$\text{Hit}_a(m^*) = \{m \in \mathcal{S} : \delta_a(\tau(m)) = \tau(m^*)\}.$$

Then

$$|\text{Hit}_a(m^*)| \leq |\Omega_a|.$$

Proof. If two distinct canonical-hit sources shared one observation o , write them as (o, T_1) and (o, T_2) . Both local updates would equal the same canonical target. Tail injectivity would force $T_1 = T_2$, hence the physical sources would coincide. Applying Θ would then make the semantic sources equal, a contradiction. Therefore every canonical-hit source requires a distinct observation. $\square$

Corollary 8.16 (Quantitative forced-debt frontier). *Let*

$$\text{Debt}_a(m^*) = \mathcal{S} \setminus \text{Hit}_a(m^*)$$

be the sources whose a -transition fails to hit the chosen canonical target. If $\mathcal{S}$ is finite, then

$$|\text{Debt}_a(m^*)| \geq |\mathcal{S}| - |\Omega_a|.$$

If $\mathcal{S}$ is infinite and Ω_a is finite, then

$$|\text{Debt}_a(m^*)| = \infty.$$

For a bounded- d top policy with finite control Q and stack alphabet Γ ,

$$|\Omega_a| \leq |Q| \sum_{j=1}^d |\Gamma|^j,$$

so the corresponding finite-source lower bound is

$$|\text{Debt}_a(m^*)| \geq |\mathcal{S}| - |Q| \sum_{j=1}^d |\Gamma|^j.$$

Thus locality can force not merely the *existence* of debt but a quantitative number of debt-producing boundary edges.

8.5 Multi-type Dyck: synthesized interior and forced harmless debt

Return to the semantic core

$$M_k = \mathcal{I}_k^* \cup \{\perp\}$$

from Example 3.14. Let

$$\mathcal{L} = \mathcal{I}_k^*$$

and use the live partial section

$$\tau_{\mathcal{L}}(s) = (\mathbf{G}, sZ),$$

with stack top on the left.

For every opening symbol o_i ,

$$(\mathbf{G}, sZ) \xrightarrow{o_i} (\mathbf{G}, isZ) = \tau_{\mathcal{L}}(is),$$

and for every matching close,

$$(\mathbf{G}, isZ) \xrightarrow{c_i} (\mathbf{G}, sZ) = \tau_{\mathcal{L}}(s).$$

Hence $\mathcal{I}_k^*$ is an exactly synthesized zero-debt canonical interior.

Now fix a closing symbol c_j . The live semantic sources that collapse to $\perp$ are

$$\mathcal{S}_j = \{\varepsilon\} \cup \{it : i \neq j, t \in \mathcal{I}_k^*\}.$$

For $k \geq 2$ this set is infinite. In the natural one-top representation, the only live boundary observations are the bottom marker Z and the $k - 1$ mismatching top types. Thus the observation capacity is finite while $\mathcal{S}_j$ is infinite. Corollary 8.16 gives:

Theorem 8.17 (Dyck forced-debt frontier). *For every $k \geq 2$ and every close type j , infinitely many live c_j -boundary sources must fail to hit any one chosen canonical representative of $\perp$ in a fixed finite bounded-local tail-injective policy.*

In the natural exact lift with

$$\tau(\perp) = (\mathbf{D}, Z),$$

underflow hits the canonical dead representative, while every nonempty mismatch

$$it, \quad i \neq j,$$

enters dead control while preserving the old stack:

$$(\mathbf{G}, itZ) \xrightarrow{c_j} (\mathbf{D}, itZ).$$

Hence every such mismatch creates representation debt.

This forced debt is nevertheless harmless. The semantic state $\perp$ is absorbing, and every physical configuration in the dead fiber rejects. Corollary 5.2 therefore gives

$$\Theta^{-1}(\perp) \subseteq \mathcal{B}.$$

We have thus obtained an exact middle regime:

$$\begin{array}{c} \text{zero-debt canonical synthesis on } \mathcal{I}_k^* \\ + \text{ infinite forced but harmless debt on the collapse frontier} \end{array}$$

For $k = 1$, by contrast, the live source set collapsing to $\perp$ under the unique close consists only of underflow:

$$\mathcal{S} = \{\varepsilon\}.$$

The natural transition reaches $(\mathbf{D}, Z)$ directly, so total canonical synthesis is possible. The one-type/multi-type phase change can therefore be stated as

$$\text{Dyck}_1 : \text{ total canonical synthesis,}$$

versus

$$\text{Dyck}_k, k \geq 2 : \text{ canonical live interior} + \text{ infinite forced harmless-debt frontier.}$$

8.6 Synthesis hierarchy

The results of this section fit one common principle:

all semantic paths to one semantic state must admit one coherent physical representative.

The implementation of that principle depends on the semantic geometry:

semantic class	synthesis mechanism
finite rooted system	explicit graph propagation
$\mathbb{Z}$ with integer translations	finite modular skeleton + unary quotient
$\mathcal{I}_k^* \cup \{\perp\}$	partial LIFO mirroring + forced-debt boundary

The section-synthesis layer therefore complements the obstruction theory. Capacity and tail-compatibility explain why a proposed representation fails; path-outcome coherence constructs a representation when one exists in the chosen template; and the forced-debt bound quantifies the boundary when total canonical synthesis is impossible.

9 Effective verification hierarchy

The preceding parts identify three increasingly precise correctness objects: a structural reset certificate, the exact physical behavioral hull, and semantic containment cores. This final theorem family asks when these objects can be computed effectively for pushdown policy classes.

The answer is intentionally layered. Ordinary pushdown saturation is sufficient whenever the relevant target set of configurations is regular. When immediate behavioral disagreement itself is regular, saturation becomes an exact behavioral oracle. Without that regularity, deterministic pushdown language equivalence remains a more general exact fallback when both the physical continuation language and the semantic specification have effective DPDA presentations.

9.1 Regular structural targets

Fix a schedule H and write

$$\text{reach}_H(c_0) = \{\delta_w^H(c_0) : w \in \Sigma^*\}$$

for its reachable physical configurations. Let $\mathcal{R}$ be a wordwise safe-reset domain, and let

$$\text{Bad}_H^{\text{str}}$$

be a set of pushdown configurations satisfying

$$U_{\mathcal{R}}(H) \subseteq \text{Bad}_H^{\text{str}}.$$

The set may be exact or conservative. In applications it is usually defined by control state, a regular stack pattern, or a finite union of such patterns.

Theorem 9.1 (Regular structural safe-reset certification). *Assume the fixed physical policy H is represented by a standard pushdown system (with C identified with its configuration universe or an effectively regular invariant policy domain) and*

$$\text{Bad}_H^{\text{str}}$$

is a regular set of pushdown configurations containing every structurally unsafe eager target.

Then the condition

$$\text{reach}_H(c_0) \cap \text{Bad}_H^{\text{str}} = \emptyset$$

is decidable by ordinary pushdown reachability saturation. Whenever it holds,

$$H \in \text{Adm}(\mathcal{E}).$$

If, in addition, the structural presentation is activation-tight so that every reachable member of $\text{Bad}_H^{\text{str}}$ is behaviorally fatal, the test is exact.

Proof. Classical pushdown reachability represents regular configuration sets by finite automata and effectively computes predecessor closures [11]. Therefore reachability of $\text{Bad}_H^{\text{str}}$ from c_0 is decidable.

If no such configuration is reachable, then no reachable eager target leaves the safe-reset certificate. Proposition 6.2 gives $H \in \text{Adm}(\mathcal{E})$. Under activation tightness, any reachable structural bad configuration supplies a fatal activation, so the converse also holds. $\square$

Remark 9.2 (Structural reachability is not automatically behavioral failure). Without activation tightness, reaching $\text{Bad}_H^{\text{str}}$ proves only that a structural certificate has been left. The target may still lie in $\mathcal{B}_H$. This is why the structural test is deliberately placed below the exact behavioral test.

9.2 Main Theorem VIII: regular immediate-disagreement saturation

For a fixed policy H , restore the policy index on the immediate terminal disagreement set:

$$\mathcal{D}_H = \{c \in C : \alpha_H(c) \neq f(\Theta(c))\}.$$

Unlike a structural unsafe set, $\mathcal{D}_H$ is behaviorally exact at the current endpoint. The difficulty is representational: the predicate $f(\Theta(c))$ may encode a nonregular stack property. Regularity of $\mathcal{D}_H$ is therefore a genuine hypothesis.

Theorem 9.3 (Regular immediate-disagreement saturation). *Assume:*

- (i) the fixed physical policy H is represented as a standard pushdown system whose policy domain C is the full pushdown configuration universe or an effectively regular invariant subset of it;
- (ii) the immediate terminal-disagreement set $\mathcal{D}_H$ is regular and effectively constructible over pushdown configurations.

Then

$$\text{Pre}_H^*(\mathcal{D}_H)$$

is regular and effectively constructible. Consequently,

$$\mathcal{B}_H = C \setminus \text{Pre}_H^*(\mathcal{D}_H)$$

is regular and effectively constructible as well. In particular, exact behavioral safety of any encoded physical configuration is decidable by regular-language membership.

Proof. The identity

$$\mathcal{B}_H = C \setminus \text{Pre}_H^*(\mathcal{D}_H)$$

is Theorem 4.5. By the classical pushdown saturation result, the predecessor closure of a regular set of configurations is regular and effectively constructible [11]. Regular configuration languages are effectively closed under Boolean operations. Since the policy domain C is regular by hypothesis,

$$C \setminus \text{Pre}_H^*(\mathcal{D}_H)$$

is therefore regular and effectively constructible. Hence $\mathcal{B}_H$ is regular and effective. $\square$

Corollary 9.4 (Exact root classification by saturation). *Under the hypotheses of Theorem 9.3,*

$$H \in \text{Adm}(\mathcal{E}) \iff c_0 \notin \text{Pre}_H^*(\mathcal{D}_H).$$

Equivalently,

$$H \notin \text{Adm}(\mathcal{E}) \iff \text{reach}_H(c_0) \cap \mathcal{D}_H \neq \emptyset.$$

Proof. Combine Corollary 4.6 with Theorem 9.3. □

Corollary 9.5 (Exact finite schedule classification in the regular subclass). *Let $\mathcal{E}$ be finite and assume every schedule has the standard pushdown-domain representation required in Theorem 9.3. If every $H \subseteq \mathcal{E}$ yields an effectively constructible regular $\mathcal{D}_H$, then the entire family*

$$\text{Adm}(\mathcal{E})$$

is decidable by applying Corollary 9.4 to each schedule.

No claim is made that enumeration over $2^{|\mathcal{E}|}$ schedules is the most efficient classification method. Causal conflict theorems from Section 7 may compress the family substantially.

9.3 Reachable containment without constructing the whole core

For a schedule H , write $\mathcal{T}_H^{\text{reach}}$ for the reachable-fiber containment core of Section 5 computed using that policy. The reachable semantic containment theorem gives

$$c_0 \in \mathcal{B}_H \iff m_0 \in \mathcal{T}_H^{\text{reach}}.$$

When $\mathcal{D}_H$ is regular, exact root membership in that semantic core can therefore be decided entirely on the physical side.

Corollary 9.6 (Reachable-containment root shortcut). *Under the hypotheses of Theorem 9.3,*

$$m_0 \in \mathcal{T}_H^{\text{reach}} \iff c_0 \notin \text{Pre}_H^*(\mathcal{D}_H).$$

Thus one can decide whether the root belongs to the reachable semantic containment core without explicitly constructing the whole set $\mathcal{T}_H^{\text{reach}}$.

Proof. Apply Theorem 5.9 and Corollary 9.4. □

Remark 9.7. This is a root-membership shortcut, not a symbolic construction of the whole semantic set $\mathcal{T}_H^{\text{reach}}$. For infinite semantic systems the latter may require additional structure on Θ and its fibers.

9.4 Finite semantic containment computation

The semantic containment core is itself elementary to compute when the semantic system is finite and its full-fiber terminal-correct set is already known.

Theorem 9.8 (Finite containment-core computation). *Assume M is finite and*

$$G_0 = \{m : \alpha(c) = f(m) \text{ for all } c \in \Theta^{-1}(m)\}$$

is available. Define

$$T_0 = G_0$$

and iteratively

$$T_{i+1} = \{m \in T_i : U_a(m) \in T_i \text{ for every } a \in \Sigma\}.$$

The sequence stabilizes at

$$\mathcal{T}.$$

Using reverse semantic edges and a deletion queue, the stabilization requires

$$O(|M||\Sigma|)$$

edge processing once G_0 is given.

Proof. The operator on the right is exactly the finite-state version of Ψ from Theorem 5.4. Repeated deletion removes precisely those states that are either initially outside G_0 or have an edge to a state already removed. The remaining set is the greatest forward-invariant subset of G_0 , namely $\mathcal{T}$. A reverse-edge queue processes each labeled semantic edge at most once. $\square$

Corollary 9.9 (Finite containment-failure certificate). *Let $n = |M|$. Then*

$$m \notin \mathcal{T} \iff \exists w \in \Sigma^* \text{ with } U_w(m) \notin G_0.$$

If $m \notin \mathcal{T}$, a shortest such word satisfies

$$|w| \leq n - 1.$$

Proof. A state is deleted exactly when some forward path reaches $M \setminus G_0$. A shortest witness path cannot repeat a semantic state before leaving G_0 , so it uses at most $n - 1$ edges. $\square$

9.5 General deterministic pushdown equivalence fallback

Regular immediate disagreement is a powerful special case, but it is not automatic. The predicate

$$\alpha_H(c) \neq f(\Theta(c))$$

may depend on a semantic projection whose stack-level fibers are not regular in the chosen coordinates.

For deterministic pushdown language specifications there is nevertheless a more general exact decision method.

Theorem 9.10 (DPDA-equivalence behavioral fallback). *Fix a deterministic pushdown policy H and a physical configuration c in its policy domain. Assume:*

- (i) *the continuation language accepted by the physical policy from c has an effective DPDA presentation;*
- (ii) *the semantic continuation language determined by $\Theta(c)$ has an effective DPDA presentation.*

Then membership

$$c \in \mathcal{B}_H$$

is decidable by deterministic pushdown language equivalence.

Proof. By definition,

$$c \in \mathcal{B}_H$$

exactly when the physical and semantic continuation languages from the two presentations are equal. Equivalence of deterministic pushdown languages is decidable by the theorem of Sénizergues [12]. $\square$

Corollary 9.11 (Exact fixed-policy root verification). *If both the physical recognizer under schedule H and the target semantic language have effective DPDA presentations, then*

$$H \in \text{Adm}(\mathcal{E})$$

is decidable even when no useful regular presentation of $\mathcal{D}_H$ is known.

Remark 9.12 (Imported machinery versus structural contribution). The saturation algorithm and DPDA-equivalence theorem are classical. The present contribution is the reduction hierarchy that specifies which exact set or language comparison encodes each notion of correctness:

regular structural bad target $\implies$ safe-reset certification
regular $\mathcal{D}_H \implies$ exact behavioral hull by saturation
effective DPDA continuation presentations $\implies$ exact language-equivalence fallback

The three columns have different hypotheses and should not be conflated.

10 Conclusion

This paper develops a general theory for exact semantic lifts under bounded-local real-time pushdown representation. Its central separation is between deterministic semantic evolution and the physical coordinates used to realize it. Making an exact projection and a section explicit turns local realizability, representation debt, behavioral safety, causal normalization schedules, section synthesis, and verification into separate mathematical questions. The resulting chain is

semantic fibers $\rightarrow$ local realizability $\rightarrow$ representation debt $\rightarrow$ behavioral containment $\rightarrow$ causal scheduling $\rightarrow$ synthesis $\rightarrow$ verification.

The theory does not claim to cover all pushdown automata. Its conclusions are relative to exact semantic lifts and stated physical policy categories: canonical section points need not be reachable, locality failure need not be behaviorally fatal, schedule admissibility need not be downward closed without robust-witness hypotheses, immediate disagreement need not be regular, and the modular construction is a synthesis theorem for a specific one-counter template rather than an unrestricted minimization or infinite-lift search result. Within those boundaries, the framework gives a common language for distinguishing semantic necessity from representation choice and for proving when local normalization is possible, forced, harmless, fatal, synthesizable, or effectively verifiable.

Statements and Declarations

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Data and Code Availability. The paper is theoretical and uses no empirical dataset. All definitions, constructions, and proofs needed for the mathematical results are contained in the manuscript. No external dataset is required to reproduce the theoretical results, and no executable code is required to establish the proofs.

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